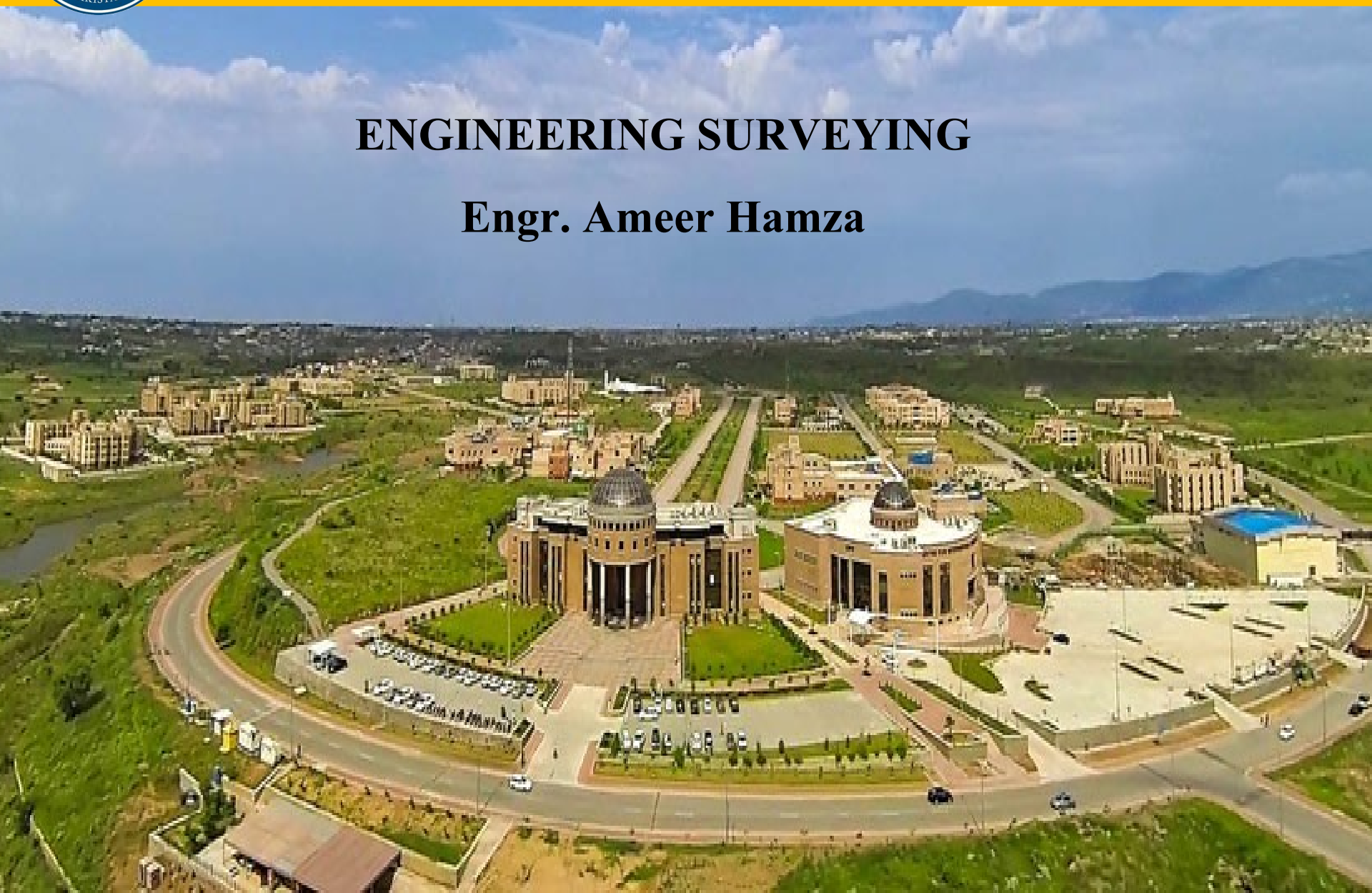

بسم الله الرحمن الرحيم

National University of Sciences and Technology
(NUST) Islamabad, Pakistan

# ENGINEERING SURVEYING

**Engr. Ameer Hamza**



# BASIC CONCEPTS
# OF
# SURVEYING AND MAPS

# SURVEY

Survey can be defined briefly as the science of measurements with the object of increasing our knowledge about the shape of earth and the features upon its surface. Mostly the end product in survey operation is a map.

# WHAT IS A MAP?

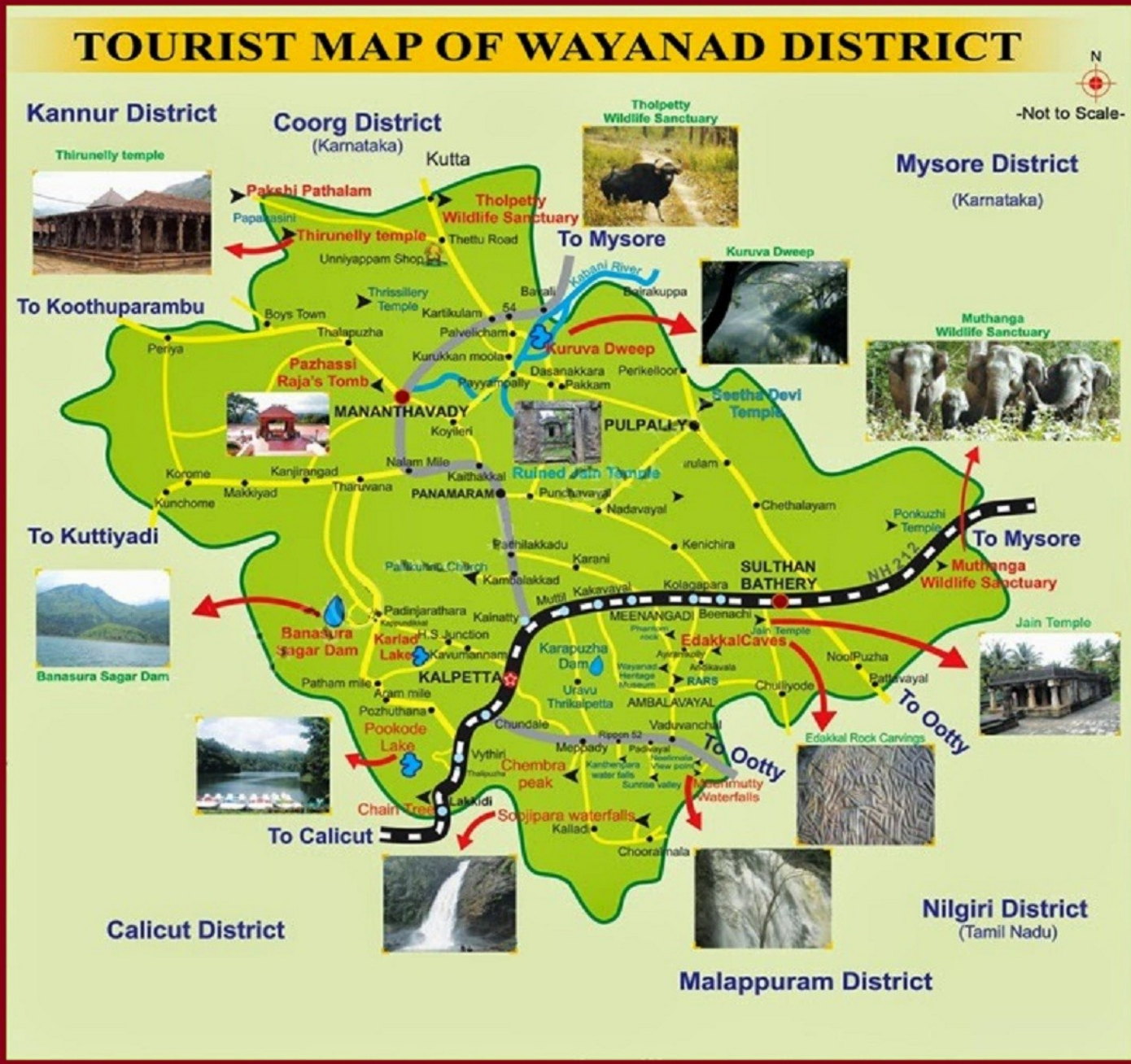
A MAP IS THE REPRESENTATION OF GROUND AND DIFFERENT FEATURES AND OTHER INFORMATION ABOUT GROUND ON A PAPER, CLOTH, PLASTIC SHEET.

# MAPS IN OUR DAILY LIFE

- TOURISM
- ROAD DISTANCE INFORMATION
- NAVIGATION
- OPERATIONAL PLANNING
- COMMUNICATION
- GEOLOGILCAL INFORMATION
- CLIMATE INFORMAION
- ADMINISTRATION
- IRRIGATION SYSTEM
- ENGINEERING PROJECTS
- CADASTRAL RECORDS
- ENVIRONMENTAL APPLICATIONS etc.

# TOURISM

## TOURIST MAP OF WAYANAD DISTRICT




## DISTANCE CHART

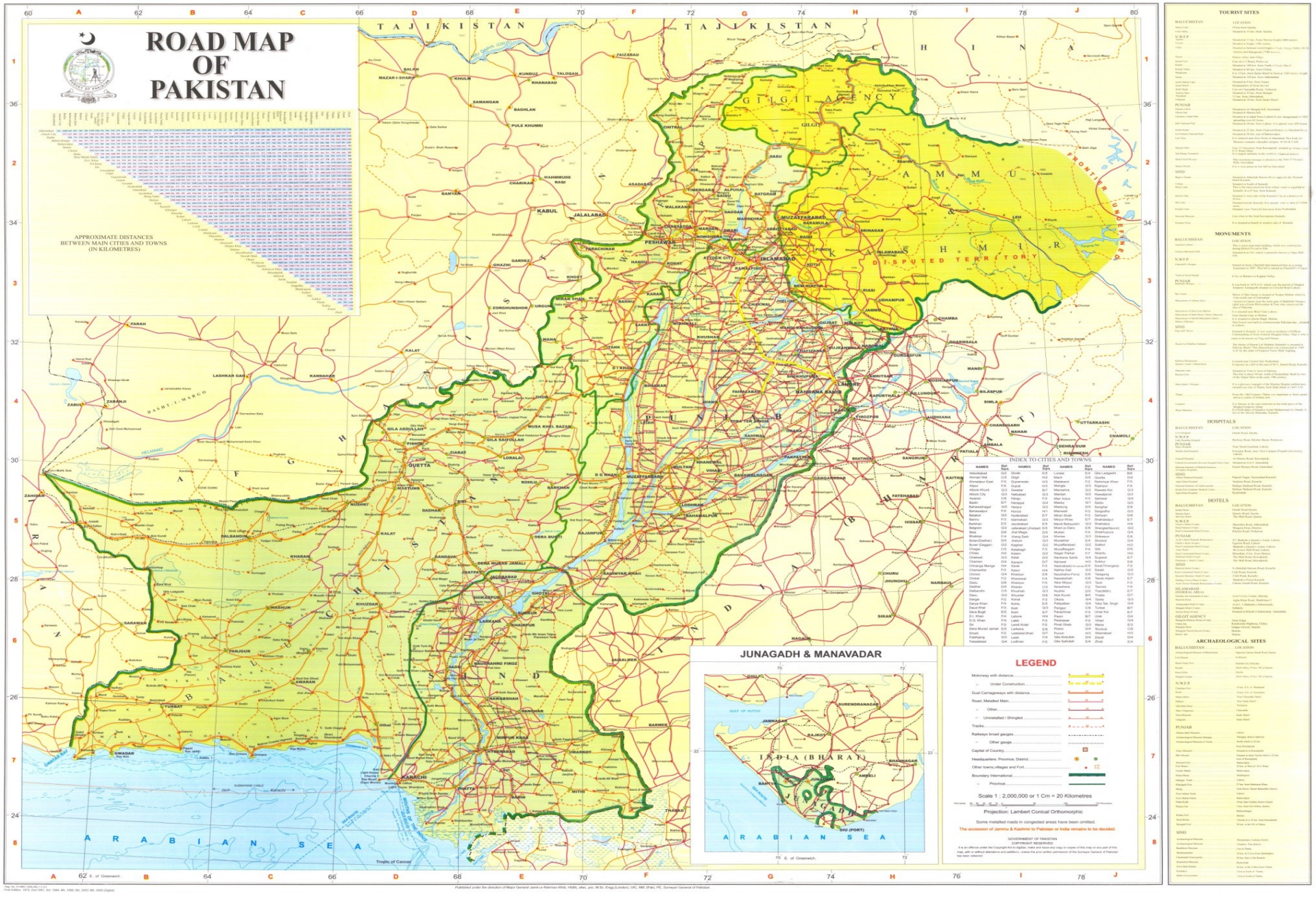
Full Form:Kalpetta, Sulthan Bathery, Mananathavady

| | KPA | SBY | MDY |
|---|---|---|---|
| Banasura Sagar Dam | 23Km | 46Km | 26Km |
| Chain Tree | 14Km | 36Km | 50Km |
| Chembra Peak | 17Km | 42Km | 52Km |
| Edakkal Cave | 28Km | 13Km | 45KM |
| Jain Temple | 24Km | 1Km | 44Km |
| Karalad Lake | 16Km | 41Km | 25Km |
| Karappuzha Dam | 16Km | 18Km | 41Km |
| Kanthanpara Water Falls | 18Km | 41Km | 57Km |
| Korome Mosque | 47Km | 52Km | 23KM |
| Kurva Island | 40Km | 50Km | 16Km |
| Lakkidi View Point | 15Km | 39Km | 50Km |
| Meenmutty Water Falls | 29Km | 28Km | 64Km |
| Muthanga Wildlife Sanctuary | 41Km | 15Km | 58Km |
| Neelimala View Point | 27Km | 25Km | 61Km |
| Pazhasi Tomb | 35Km | 44Km | 1Km |
| Pakshpathalam | 70Km | 75Km | 35Km |
| Soojipara Water Falls | 22Km | 46Km | 58Km |
| Sunrise Valley | 22Km | 23Km | 57Km |
| Thirunelly Temple | 64Km | 70Km | 31Km |
| Tholpetty Wildlife Sanctuary | 55Km | 65Km | 25Km |

## ENTRY TIME

| | |
|---|---|
| Pookode Lake | : 9.00 Am - 5.30 Pm |
| Soojipara Water Falls | : 8.00 Am - 5.00 Pm |
| Chembrapeak (trekking) | : 7.00 Am - 2.00 Pm |
| (visiting) | : 7.00 Am - 5.00 Pm |
| Meenmutty Waterfalls | : 7.00 Am - 4.00 Pm |
| Banasura Sagardam | : 8.30 Am - 5.30 Pm |
| Karalad Lake | : 9.00 Am - 5.00 Pm |
| Edakkal Cave | : 9.00 Am - 4.00 Pm |
| Kuruva Island | : 9.00 Am - 3.30 Pm |
| Pazhassi Tomb | : 10.00Am - 5.00 Pm |
| Muthanga Wildlife Sanctuary | : 7.00 Am - 10.00Am |
| | 3.00 Pm - 5.00 Pm |
| Tholpetty Wildlife Sanctuary | : 7.00 Am - 10.00Am |
| | 3.00 Pm - 5.00 Pm |
| Jain Temple Bathery | : 8.00 Am - 12.00Am |
| | 2.00 Pm - 6.00 Pm |
| Ambalavayal Museum | : 9.00 Am - 6.00 Pm |

# ROAD DISTANCE INFORMATION



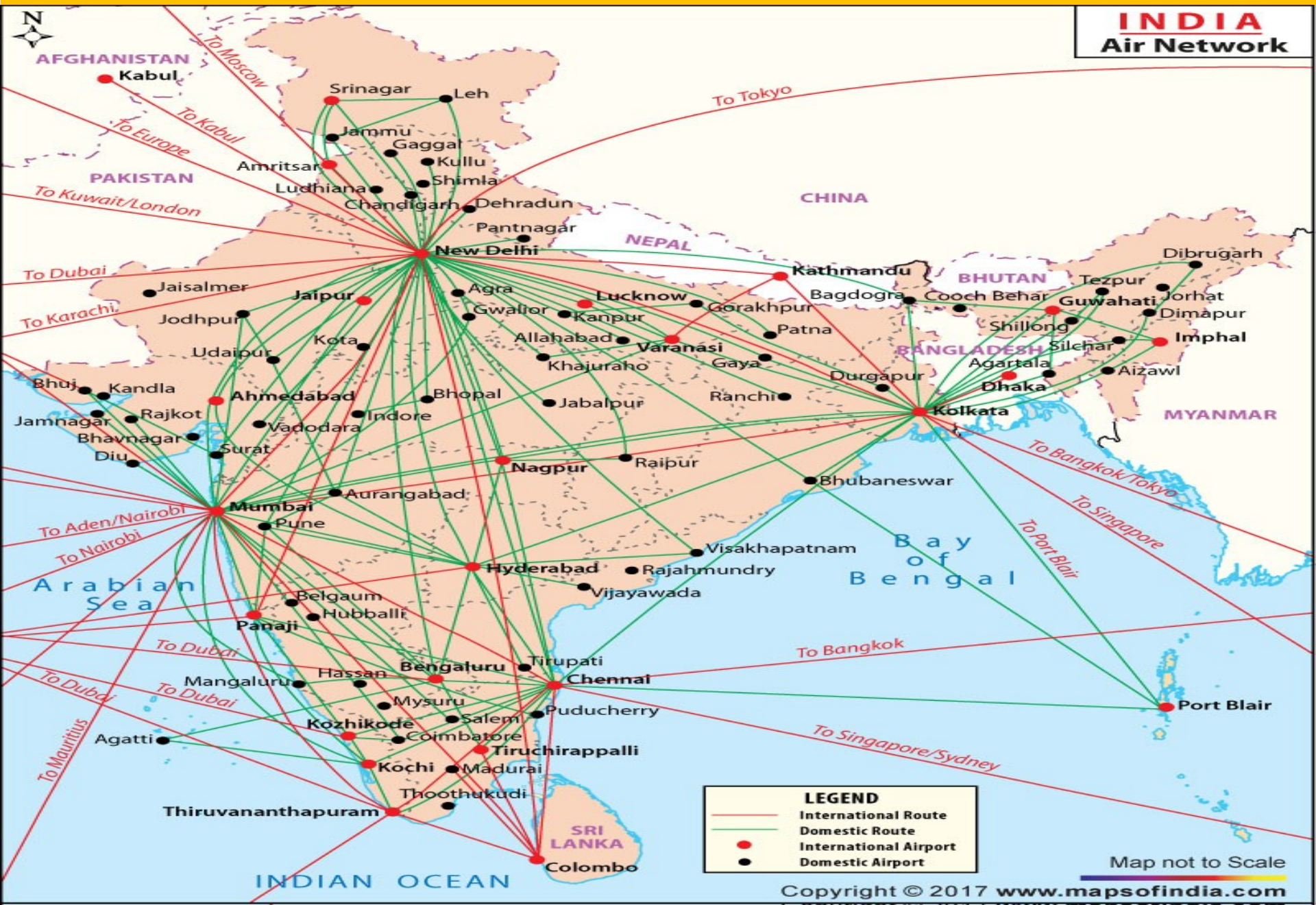
# NAVIGATION


iPad
1:44 PM
92%
Rye Town Park
Marshland Park
Park
Rye Nature Ctr
Edgewood Park
69 mi
SPEED LIMIT 55
65
+58 mins.
17.2 mi
Exit 6A - 95 North - Cross Bronx Expressway
continue for 6.1 mi
Take Exit 6A to 95 North - Cross Bronx Expressway
continue for 5.4 miles
Stay on 95 South - Cross Bronx Expressway as it bea..
continue for 0.7 miles
Take George Washington Bridge Upper Level - 95 ...
continue for 4.4 miles
Take left Exit 69 to 80 West - Express
continue for 396 miles
Take ramp to 80 West - Ohio Turnpike
continue for 354 miles
Take Exit 21 to Ramp to 80 West
continue for 23.8 miles
2,816 mi
4:51 am

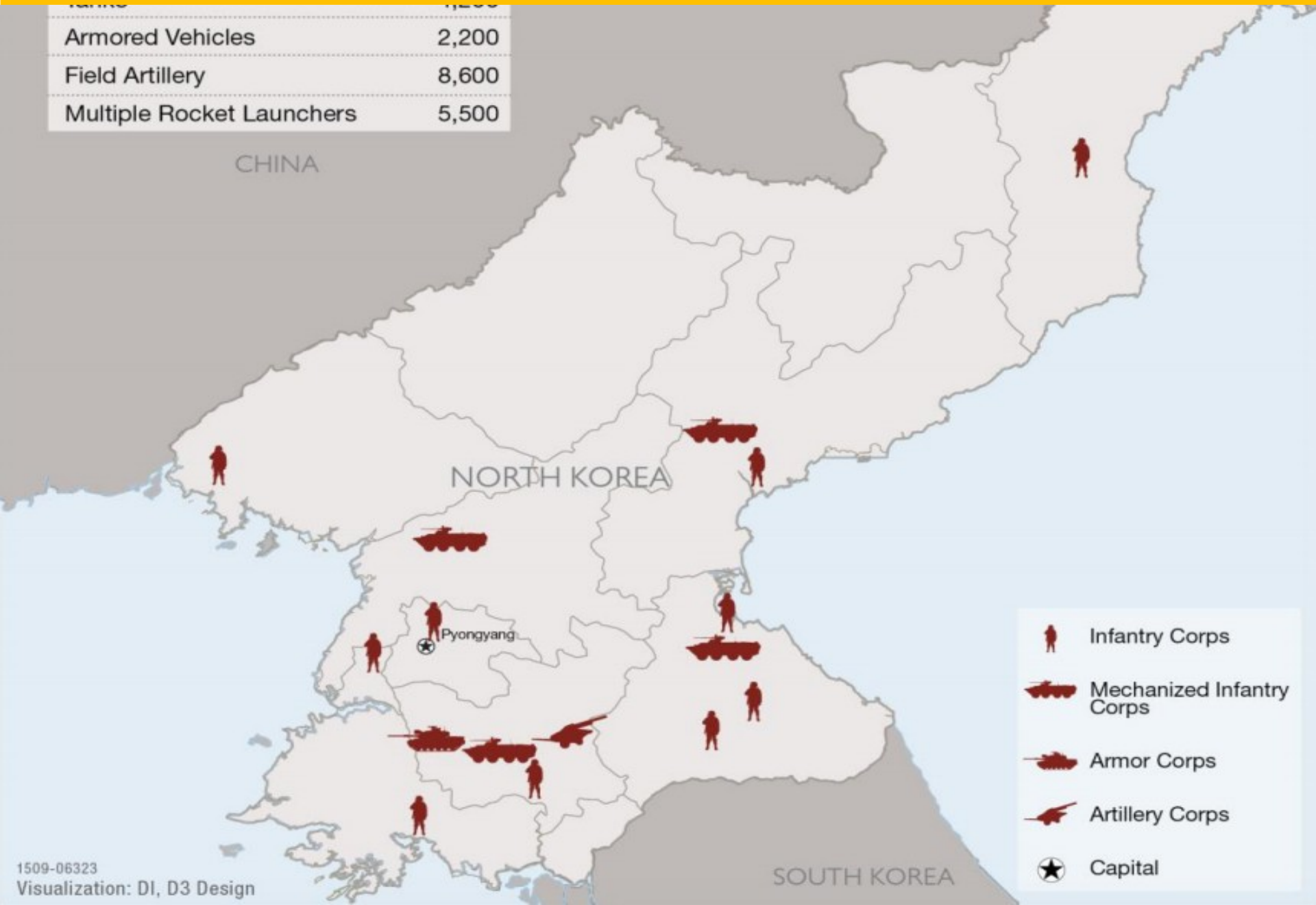
# COMMUNICATION MAP



# COMMUNICATION MAP



# OPERATIONAL PLANNING



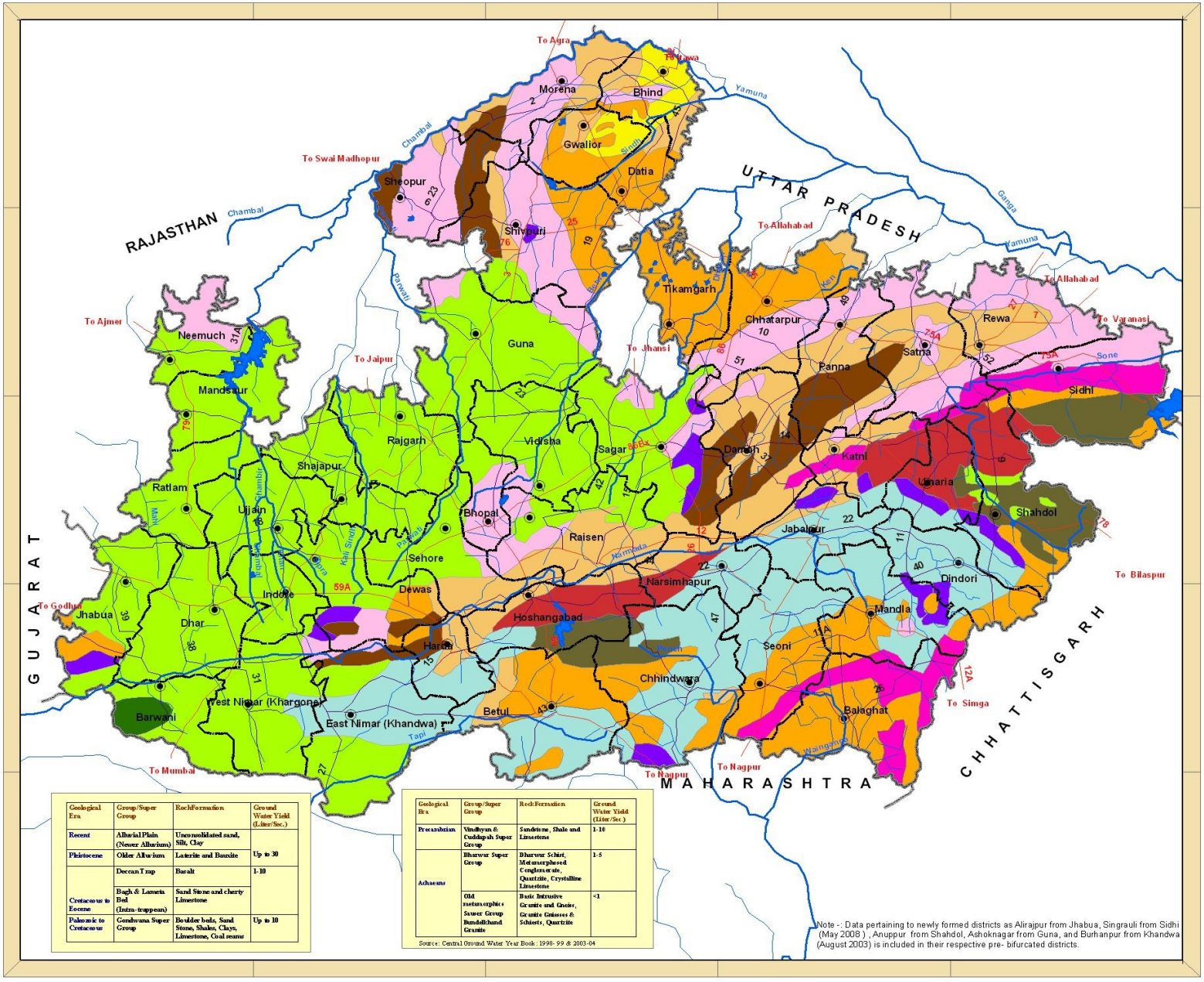
# OPERATIONAL PLANNING



# OPERATIONAL PLANNING



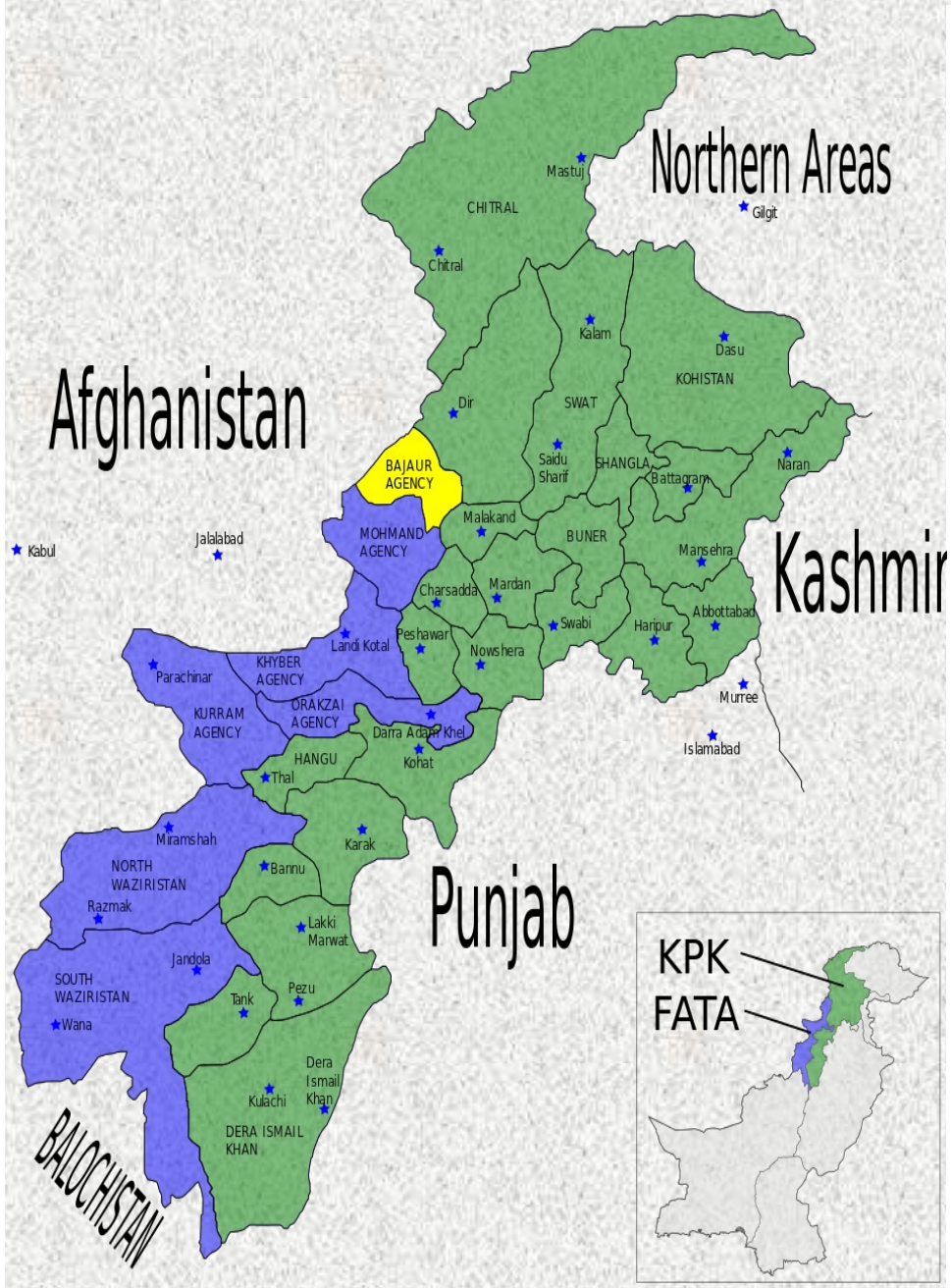
# GEOLOGILCAL INFORMATION



# CLAMATE INFORMAION



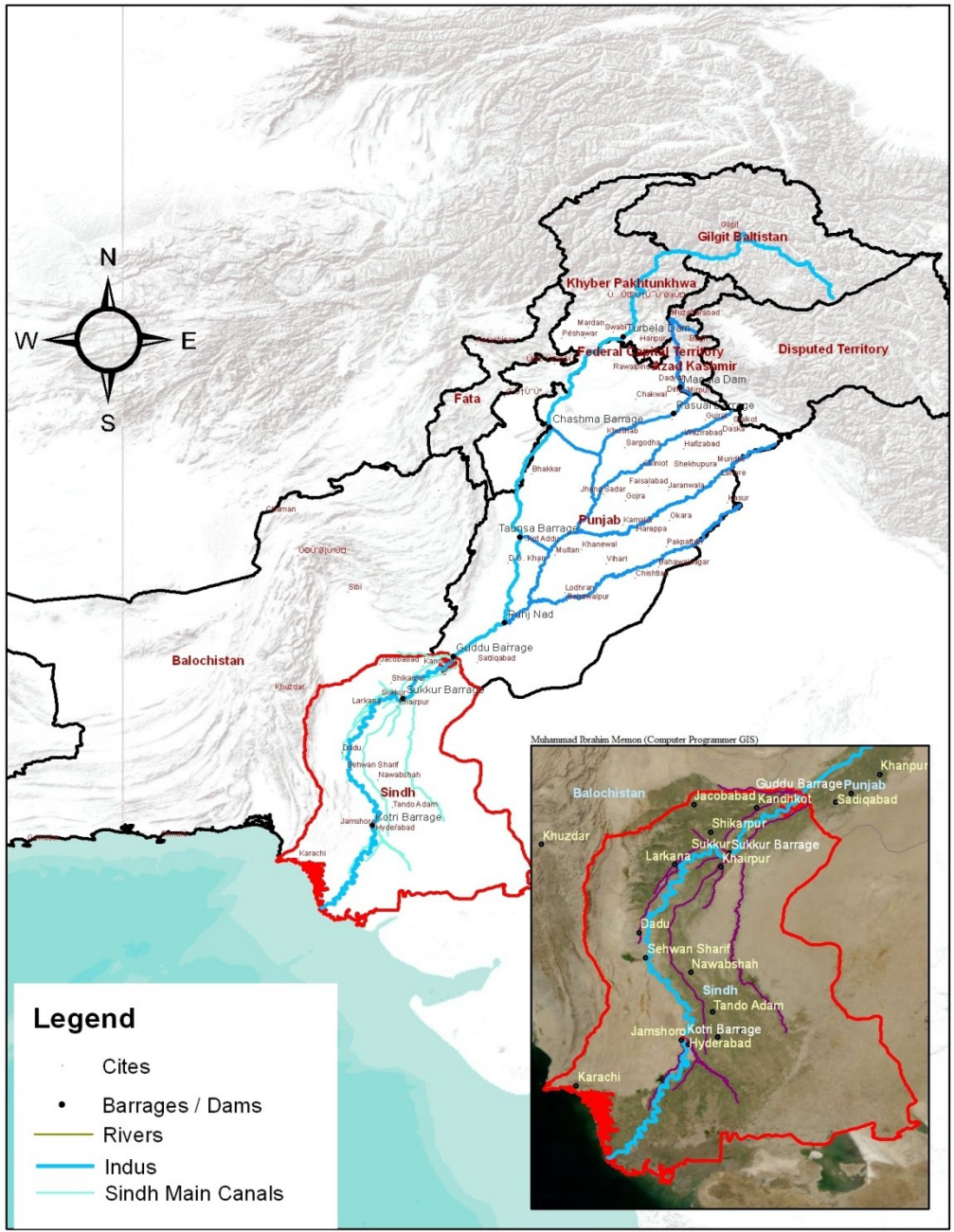
# ADMINISTRATION

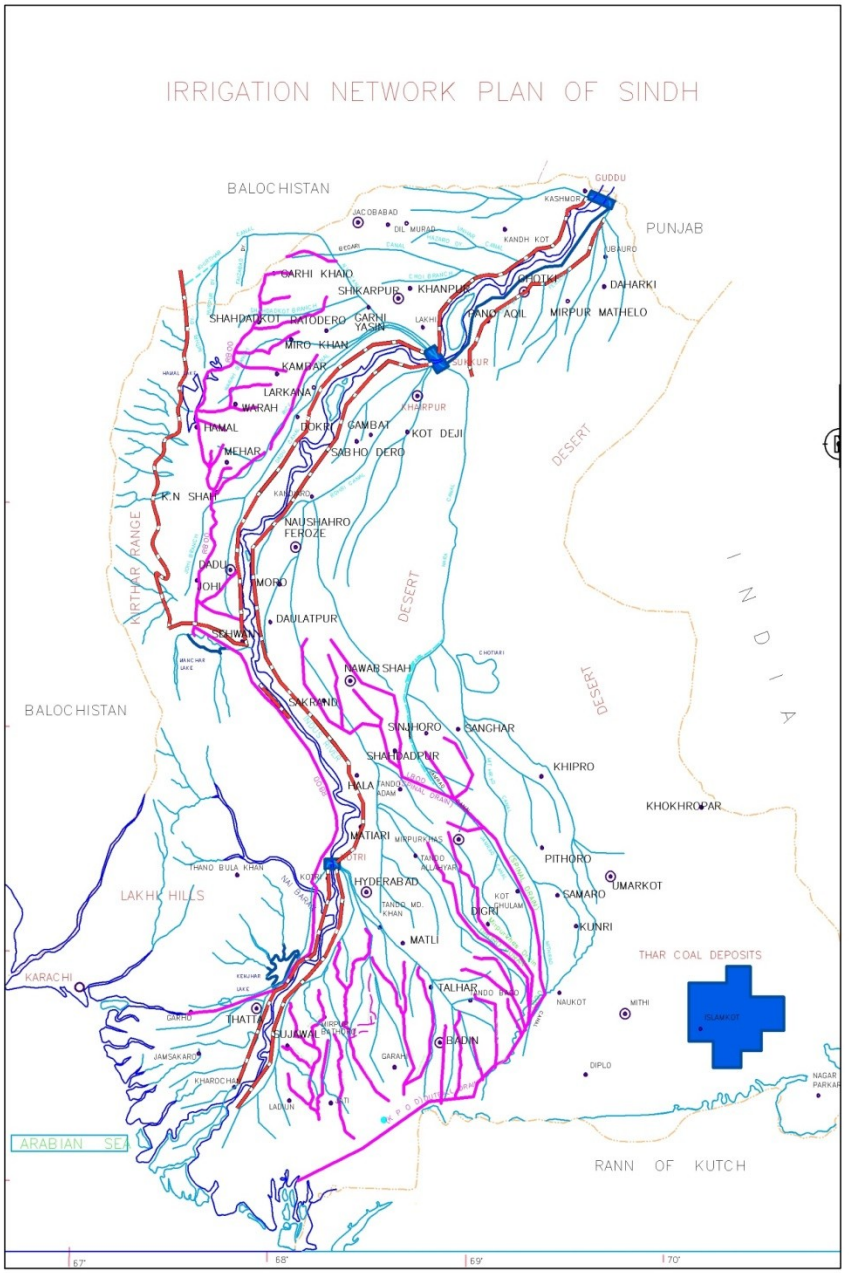




Administrative Map of Kenya
TURKANA
MARSABIT
MOYALE
MANDERA
WAJIR
SAMBURU
ISIOLO
BARINGO
WEST POKOT
TRANS NZOIA
UASIN GISHU
ELGEYO MARAKWET
BUNGOMA
BUSIA
SIAYA
NANDI
LAIKIPIA
KOIBATEK
MERU NORTH
MERU CENTRAL
GARISSA
KERICHO
NAKURU
NYANDARUA
NYERI
THARAKA
EMBU
BURET
BOMET
HOMA BAY
MURANGA
KIRINYAGA
MBEERE
MWINGI
MIGORI
KURIA
NAROK
THIKA
MACHAKOS
KITUI
TANA RIVER
IJARA
KAJIADO
MAKUENI
LAMU
TAITA TAVETA
MALINDI
KILIFI
KWALE
MOMBASA
Lodwar
Lokichar
Lokichokio
Kakuma
Lo Kitaung
Saberei
North Horr
South Horr
Baragoi
Laisamis
Kargi
Choba
Marsabit
Moyale
Buna
El Wak
Malka Mari
Ramu
Mandera
Wajir
Tarbaj
Habaswein
Mado Gashi
Garba Tula
Merti
Sericho
Archers Post
Isiolo
Maralal
Wamba
Rumuruti
Kora Wells
Daadab
Garissa
Kapenguria
Sigor
Karpedo
Kitale
Kabarnet
Marigat
Eldama Ravine
Nakuru
Nyeri
Kerugoya
Kutus
Ngomeni
Mwingi
Kitui
Endau
Masabubu
Ijara
Hola
Wenje
Tulu
Narok
Naro Sura
Ngong
Kajiado
Ilbisil
Makindu
Mtito Andei
Loitokitok
Taveta
Voi
Mwatate
Kilifi
Mombasa
Kisumu
Kisii
Homa Bay
Migori
Kericho
Bomet
Nyamira
Kisii Gucha
Trans Mara
Kuria
Nyando
Muhoroni
Nandi Hills
Eldoret
Kitale
Lugari
Mumias
Busia
Siaya
Bondo
Nyando
Kakamega
Butere
Webuye
Vihiga
Kisumu
Nyando
Nakuru
Kericho
Buret
Bomet
Nyeri
Kerugoya
Kutus
Embu
Muranga
Kirinyaga
Mbeere
Thika
Kiambu
Nairobi
Kanguniu
Machakos
Athi River
Machakos
Kajiado
Makueni
Kitui
Mwingi
Ngomeni
Garissa
Masabubu
Ijara
Hola
Wenje
Tulu
Lamu
Malindi
Kilifi
Mombasa
Kwale
Taveta
Voi
Mwatate
Mtito Andei
Loitokitok
Makindu
Ilbisil
Kajiado
Namanga
Narok
Naro Sura
Ngong
Kitui
Endau

# IRRIGATION SYSTEM

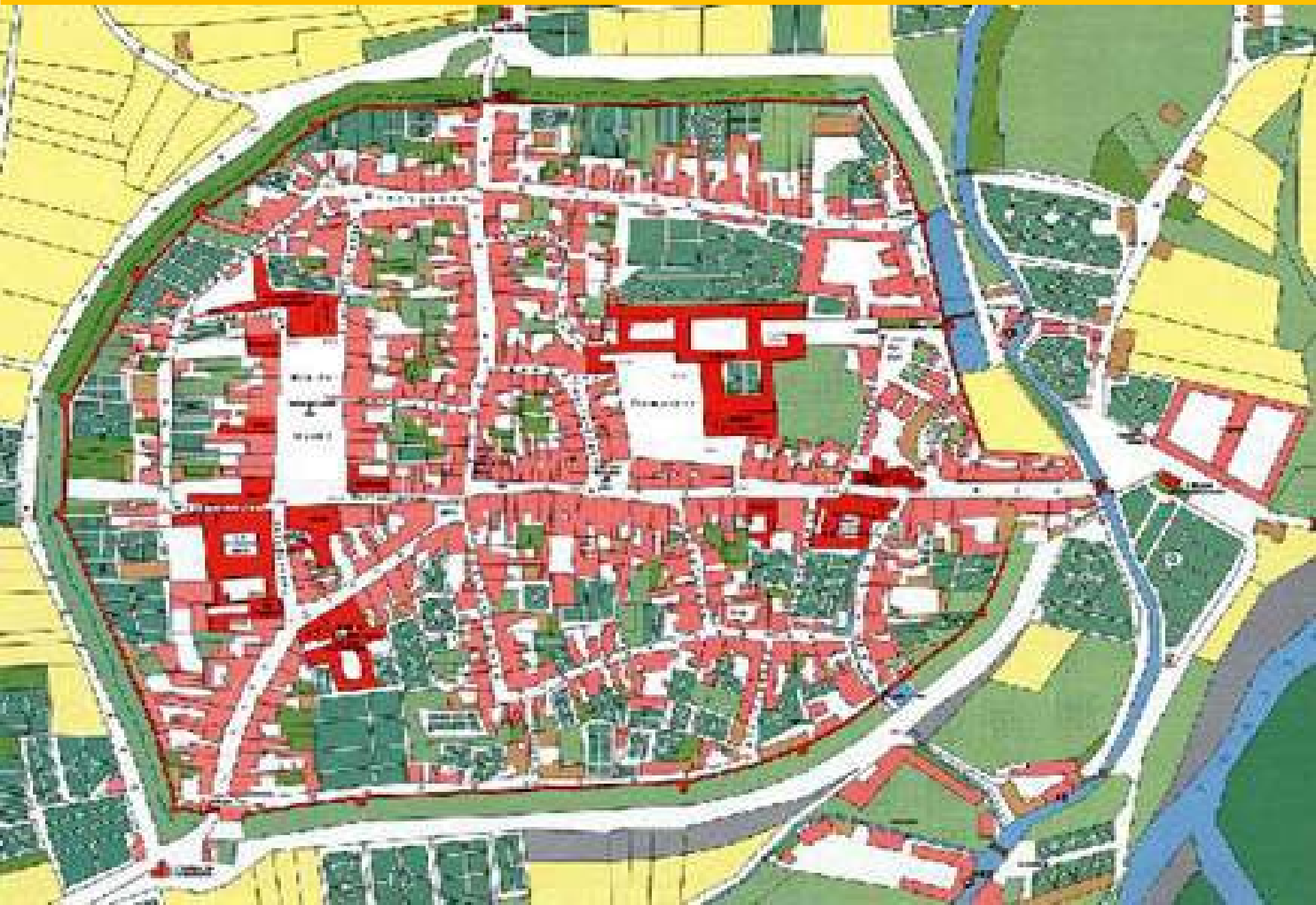


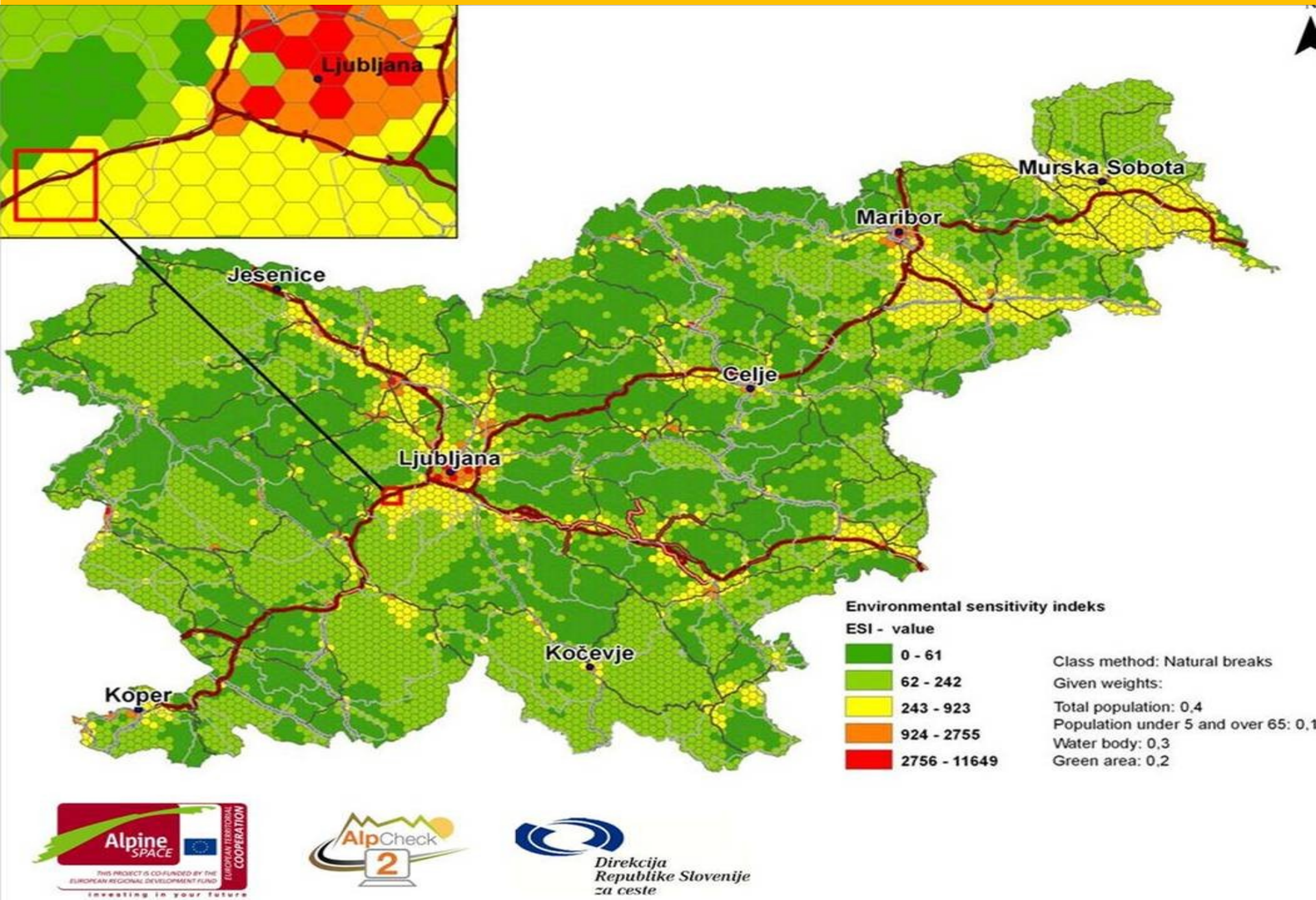
# ENGINEERING PROJECTS



# CADASTRAL RECORDS



# ENVIRONMENTAL APPLICATIONS




Alpine SPACE
EUROPEAN TERRITORIAL COOPERATION
THIS PROJECT IS CO-FUNDED BY THE EUROPEAN REGIONAL DEVELOPMENT FUND
investing in your future

AlpCheck 2

Direkcija Republike Slovenije za ceste

# TYPES OF MAPS

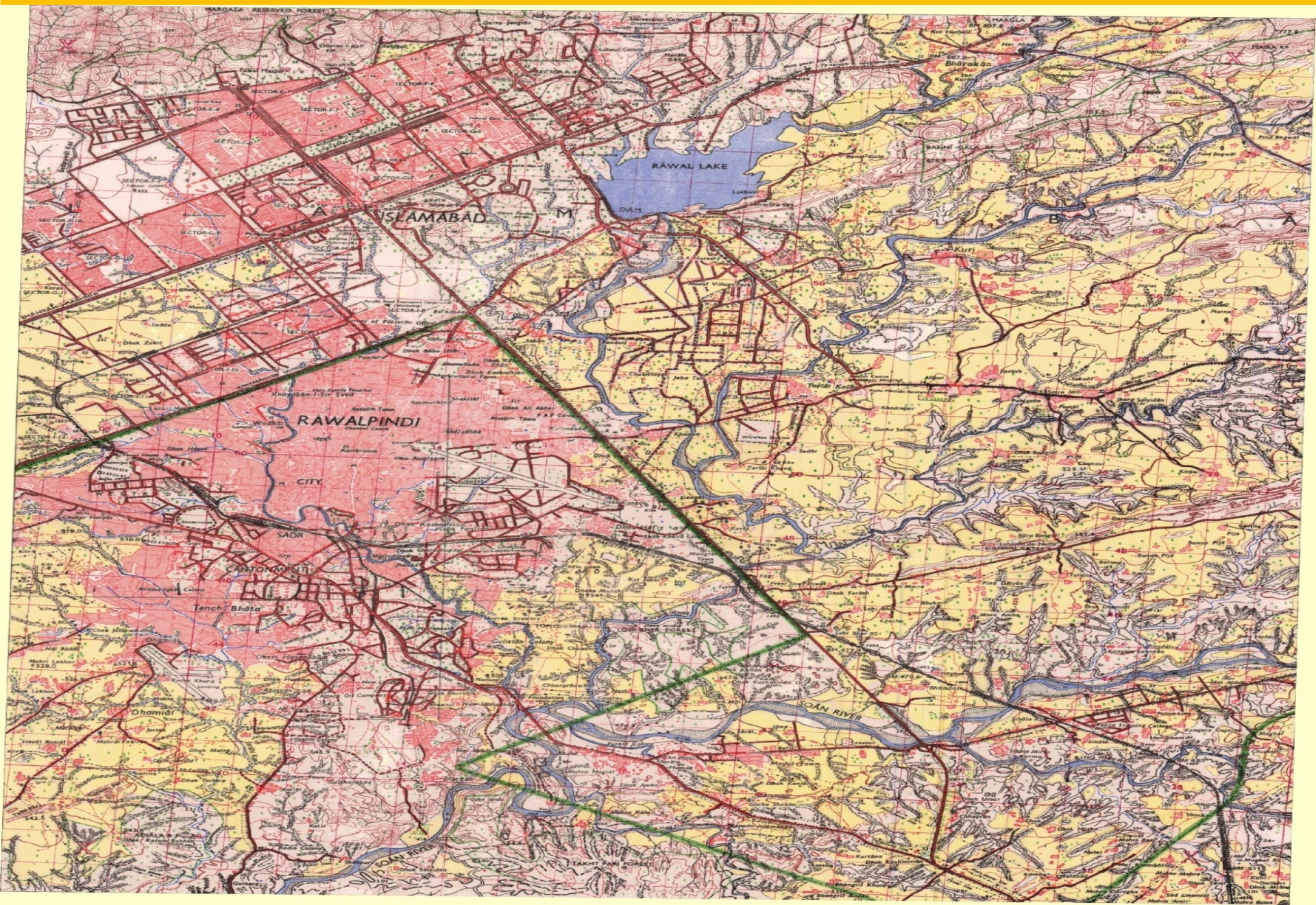
Maps are designed and made for different uses. Types depend on the nature of use. Some types are given as follows;

- Topographic Maps
  - Small Scale Maps
  - Large Scale Maps
- Photo Maps
- Relief Maps

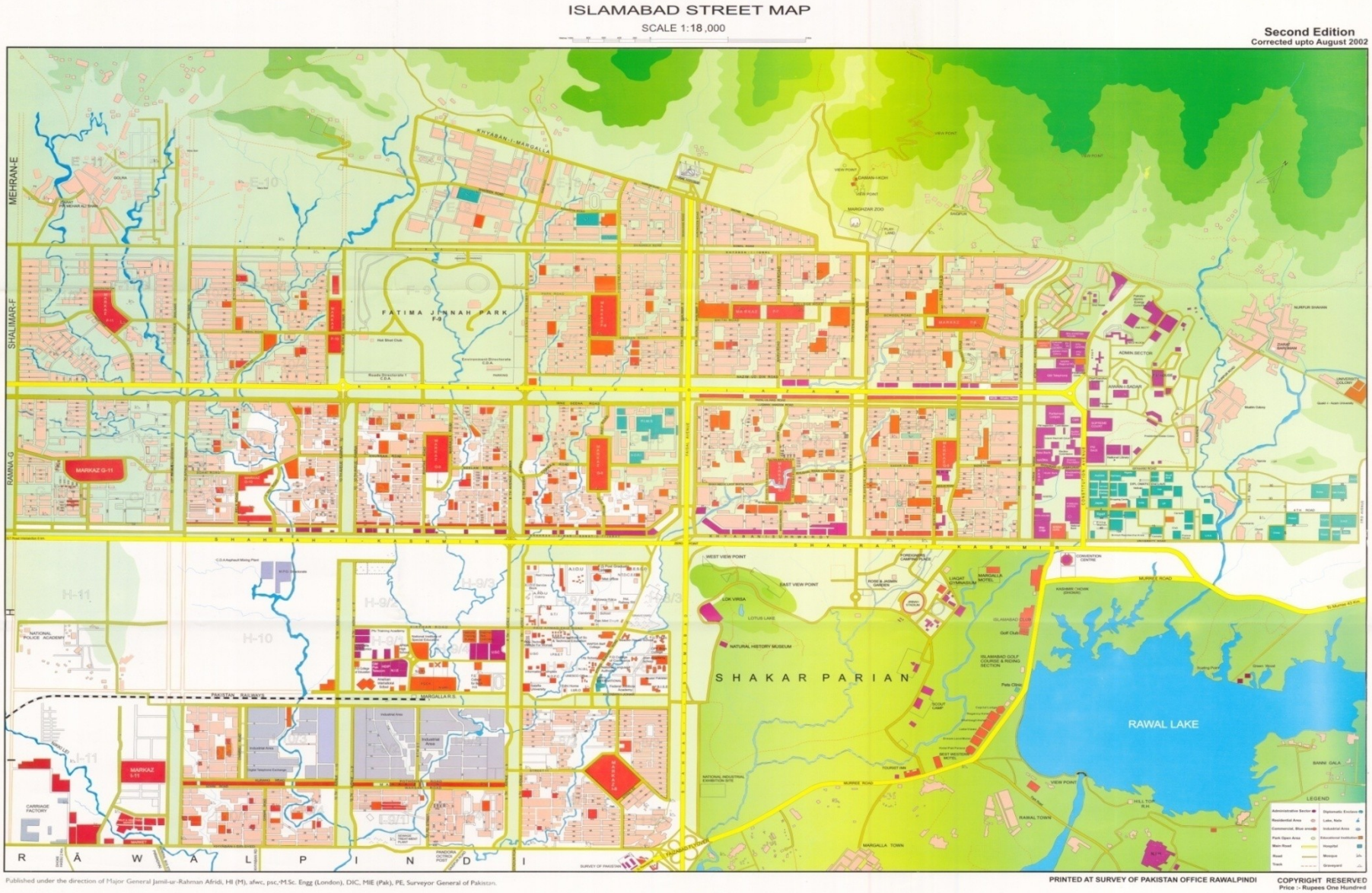
# TOPOGRAPHICAL MAPS

The topographical map is the one which shows all the information about natural and artificial objects. Scales vary between 1/5000 to 1/1000000 depending on the purpose for which the map is made.

# TOPOGRAPHICAL MAPS



# TOPOGRAPHICAL MAPS



# MAP'S SCALE

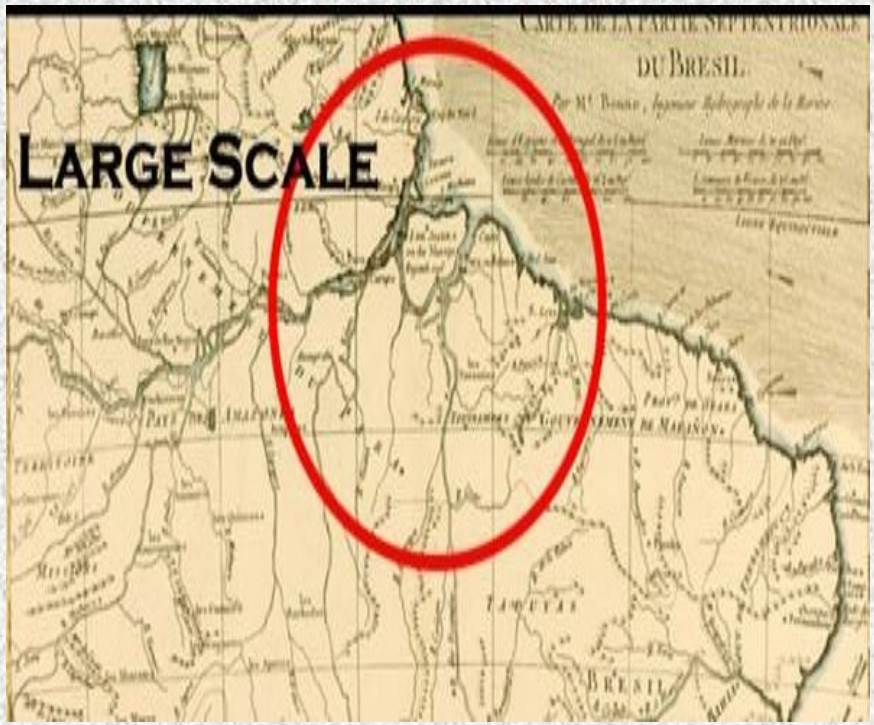
**Small Scale Maps**: These maps show large area with small details.



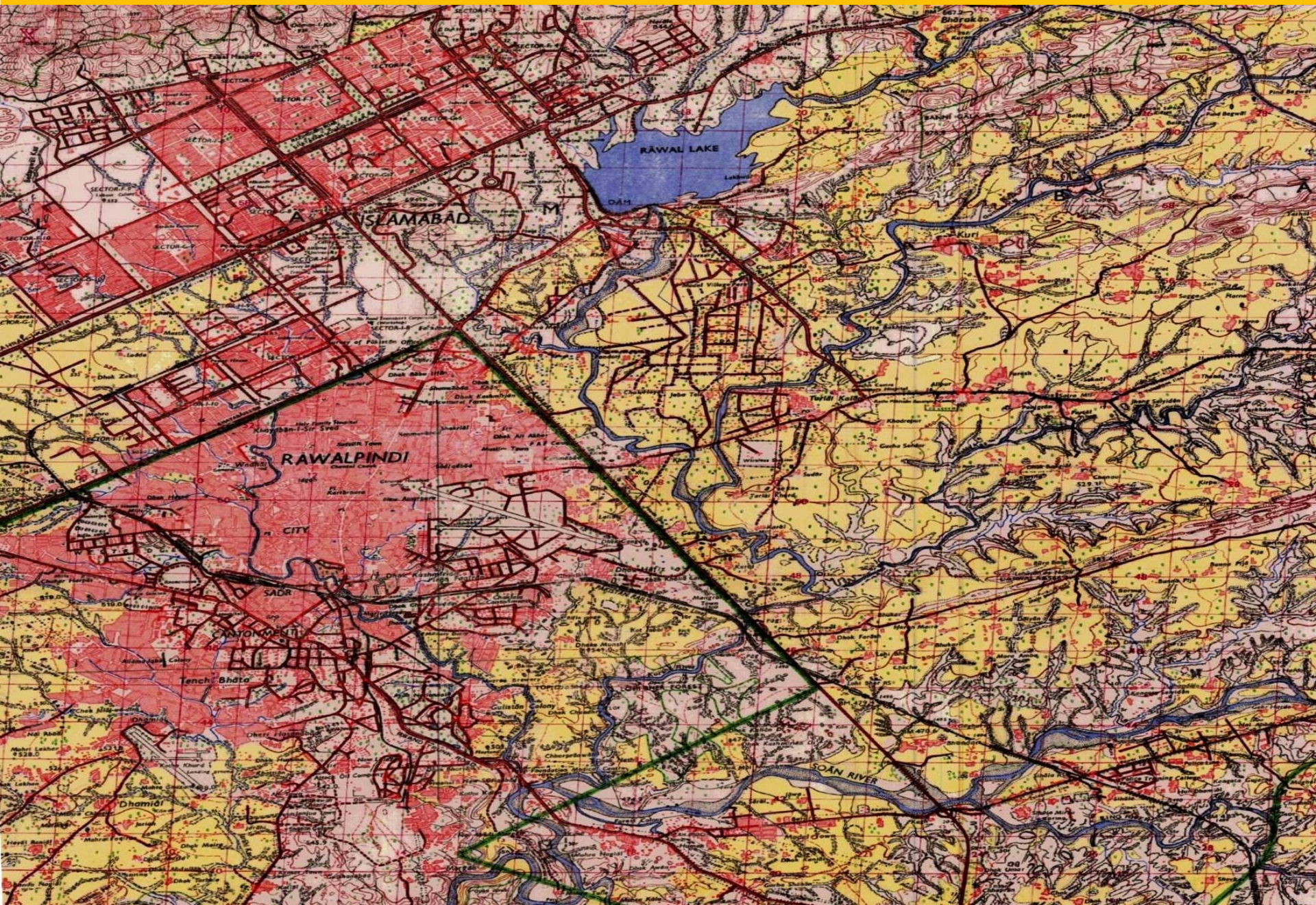

**Large Scale Maps**: These maps show small area with greater details. Some of the uses of Large-Scale Maps are as follows;

Engineering Applications.

- Cadastral Purposes.
- Any other special uses where greater details are required.



# ISLAMABAD MAP



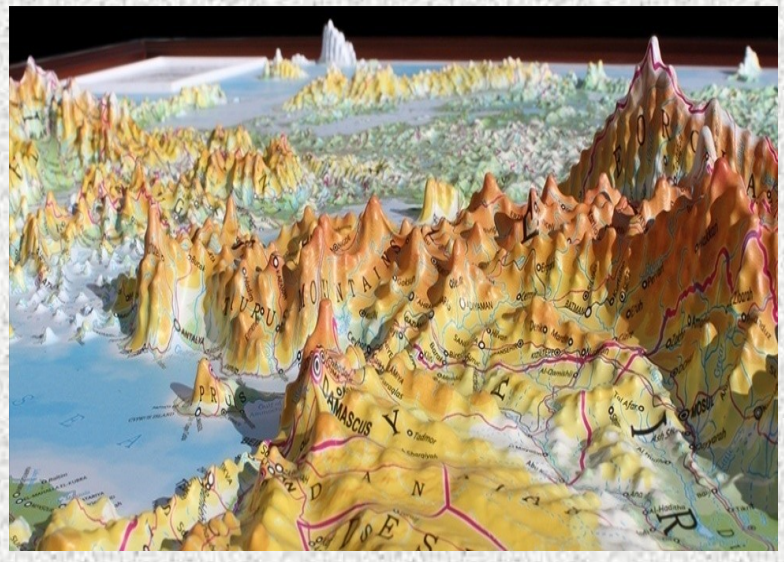
- **PHOTO MAPS**

These are made by mosaics of aerial photographs and by satellite images.



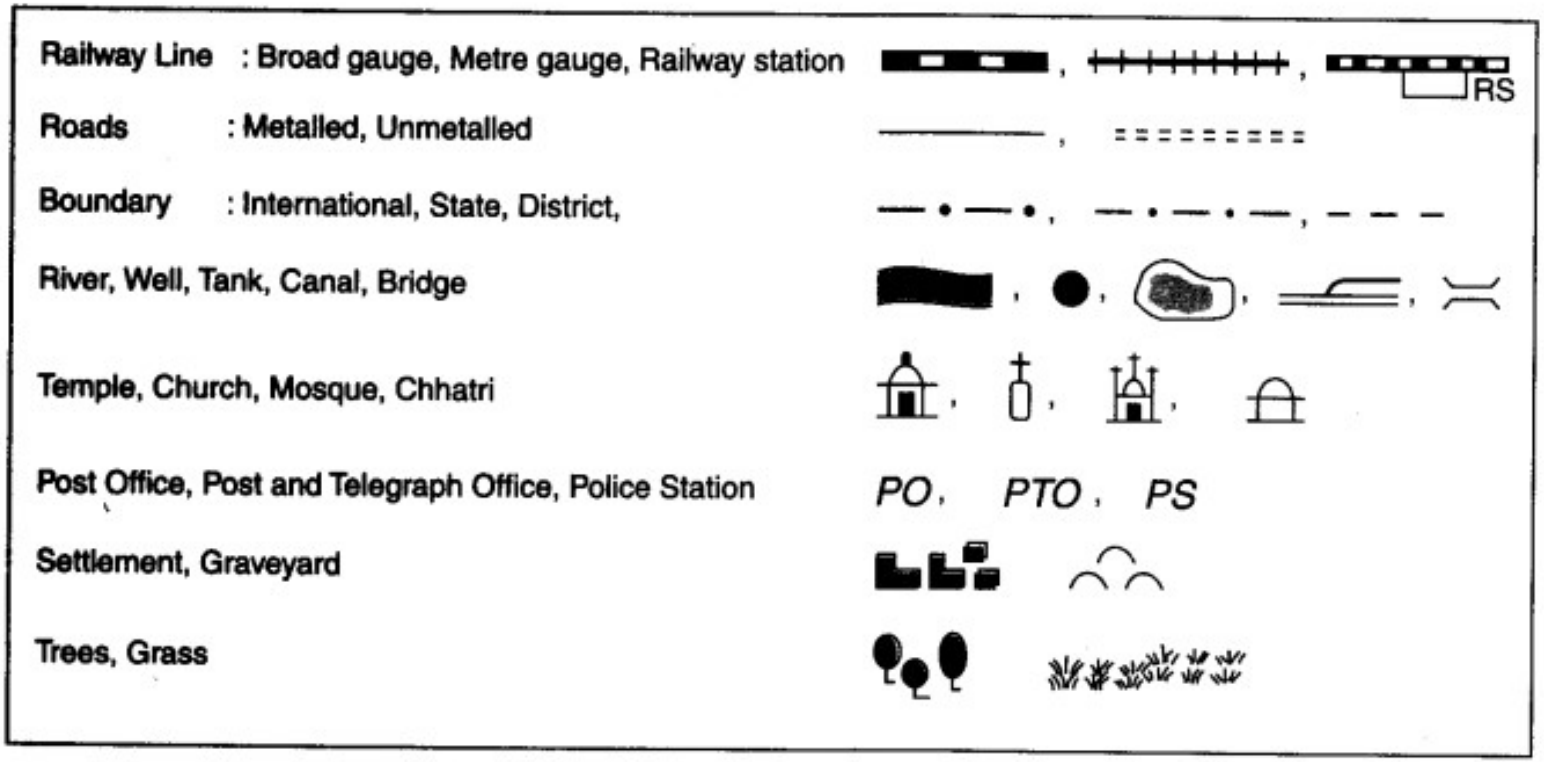

- **RELIEF MAPS**

These are 3-D maps made using different mediums like plastic etc.



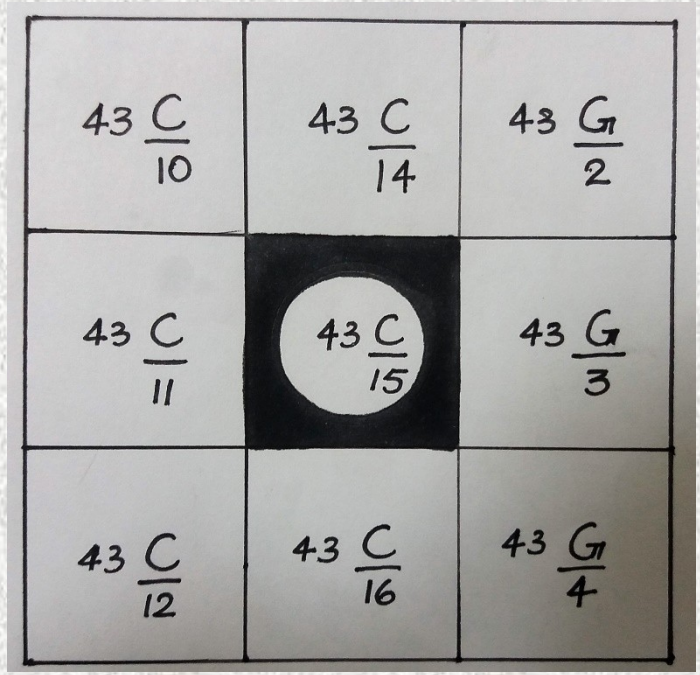
# Sign Convention

Different symbols have been designed to show different ground objects on maps. These symbols are listed on the bottom of every map in the form of a key or legend for guidance of users of maps.

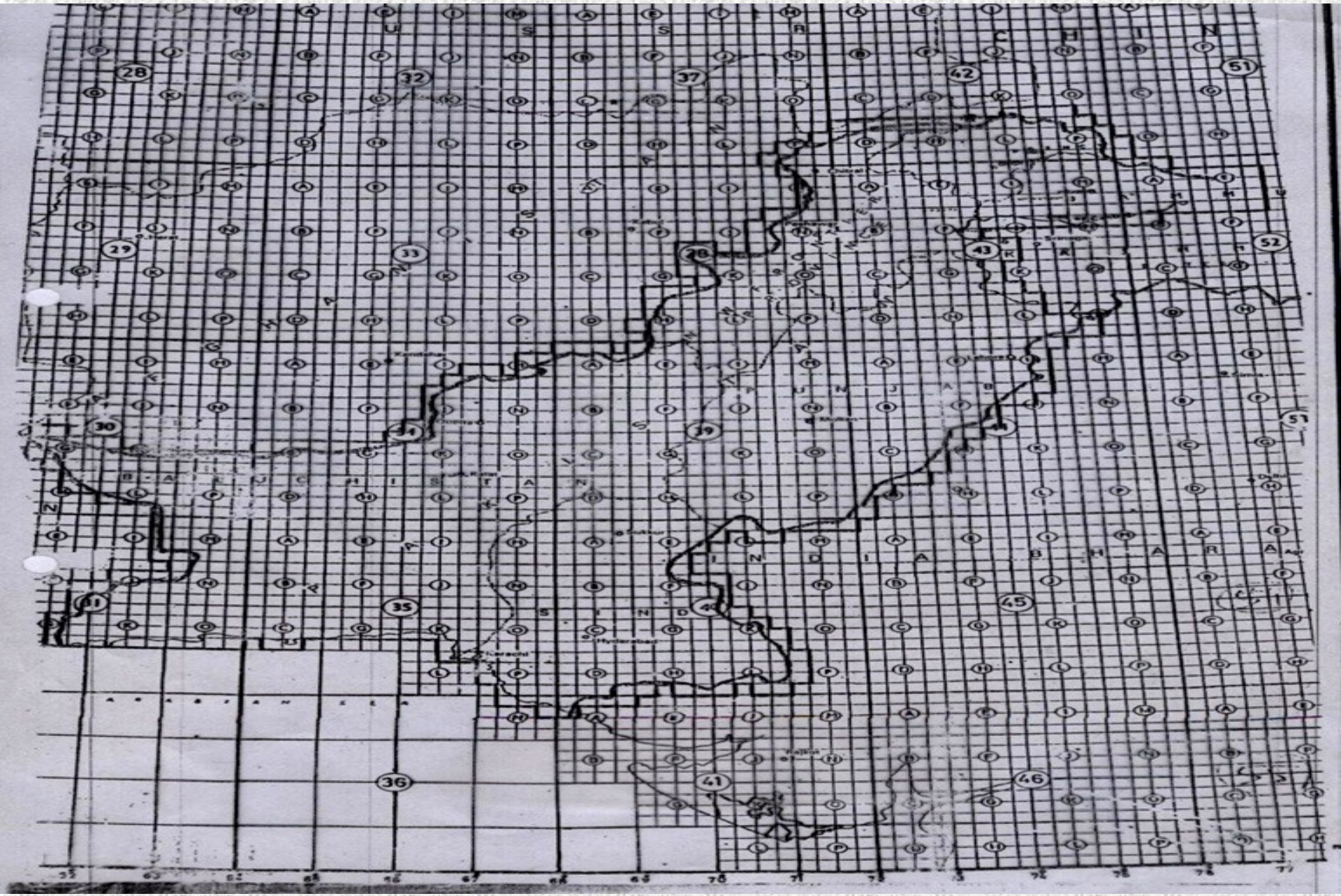


# INDEX TO MAP

All maps have been numbered on a regular system. At the bottom of each map the index shows the numbering of adjacent map sheets so that the users can conveniently demand the adjacent map sheets in case he has to work in that area.

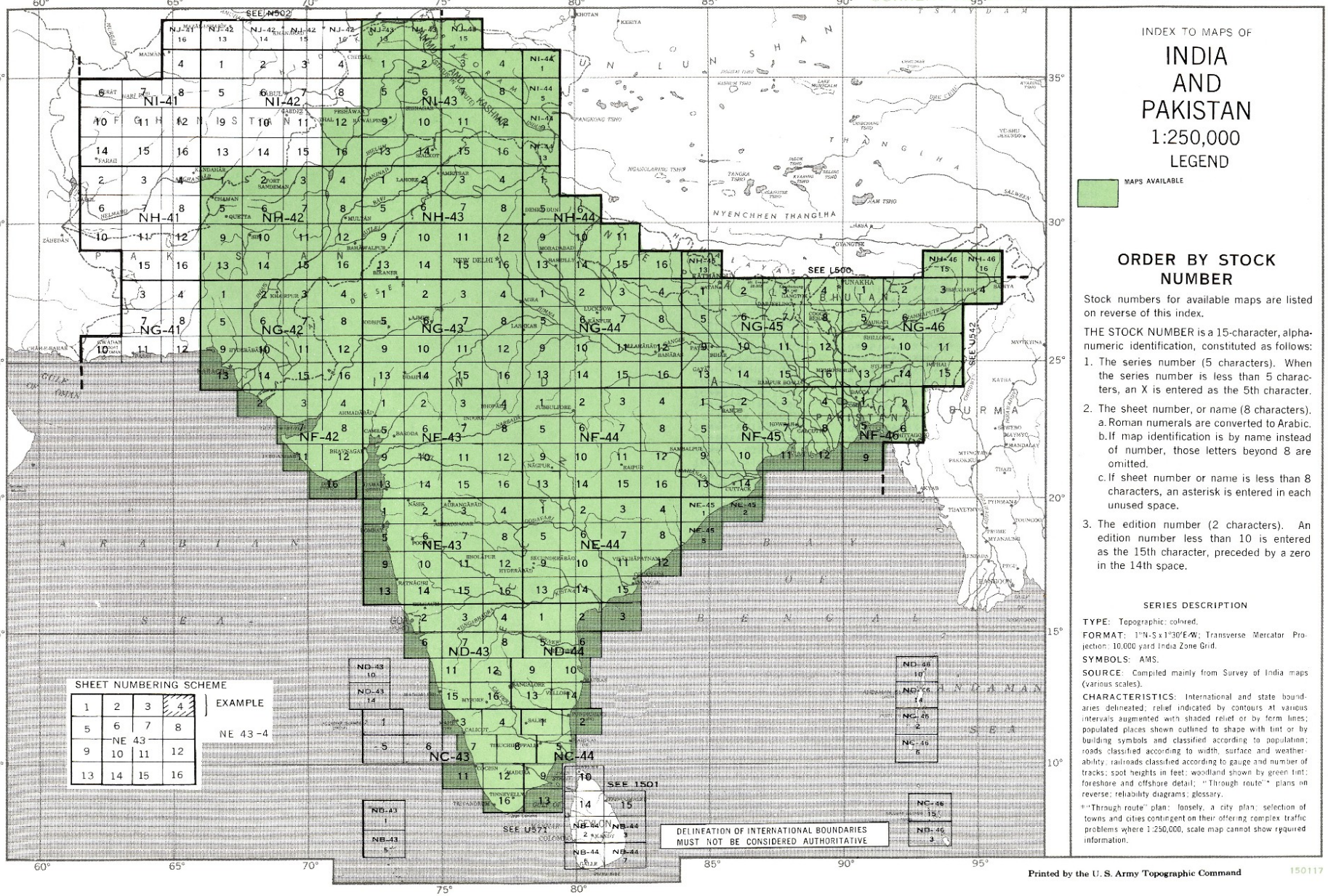


# INDEX TO MAP



# INDEX TO MAP






INDEX TO MAPS OF

# INDIA AND PAKISTAN

1:250,000

LEGEND

MAPS AVAILABLE

## ORDER BY STOCK NUMBER

Stock numbers for available maps are listed on reverse of this index.

THE STOCK NUMBER is a 15-character, alpha-numeric identification, constituted as follows:

1. The series number (5 characters). When the series number is less than 5 characters, an X is entered as the 5th character.
2. The sheet number, or name (8 characters).
   a. Roman numerals are converted to Arabic.
   b. If map identification is by name instead of number, those letters beyond 8 are omitted.
   c. If sheet number or name is less than 8 characters, an asterisk is entered in each unused space.
3. The edition number (2 characters). An edition number less than 10 is entered as the 15th character, preceded by a zero in the 14th space.

### SERIES DESCRIPTION

**TYPE:** Topographic; colored.

**FORMAT:** 1°N-S x 1°30'E-W; Transverse Mercator Projection; 10,000 yard India Zone Grid.

**SYMBOLS:** AMS.

**SOURCE:** Compiled mainly from Survey of India maps (various scales).

**CHARACTERISTICS:** International and state boundaries delineated; relief indicated by contours at various intervals augmented with shaded relief or by form lines; populated places shown outlined to shape with tint or by building symbols and classified according to population; roads classified according to width, surface and weather-ability; railroads classified according to gauge and number of tracks; spot heights in feet; woodland shown by green tint; foreshore and offshore detail; "Through route" plans on reverse; reliability diagrams; glossary.

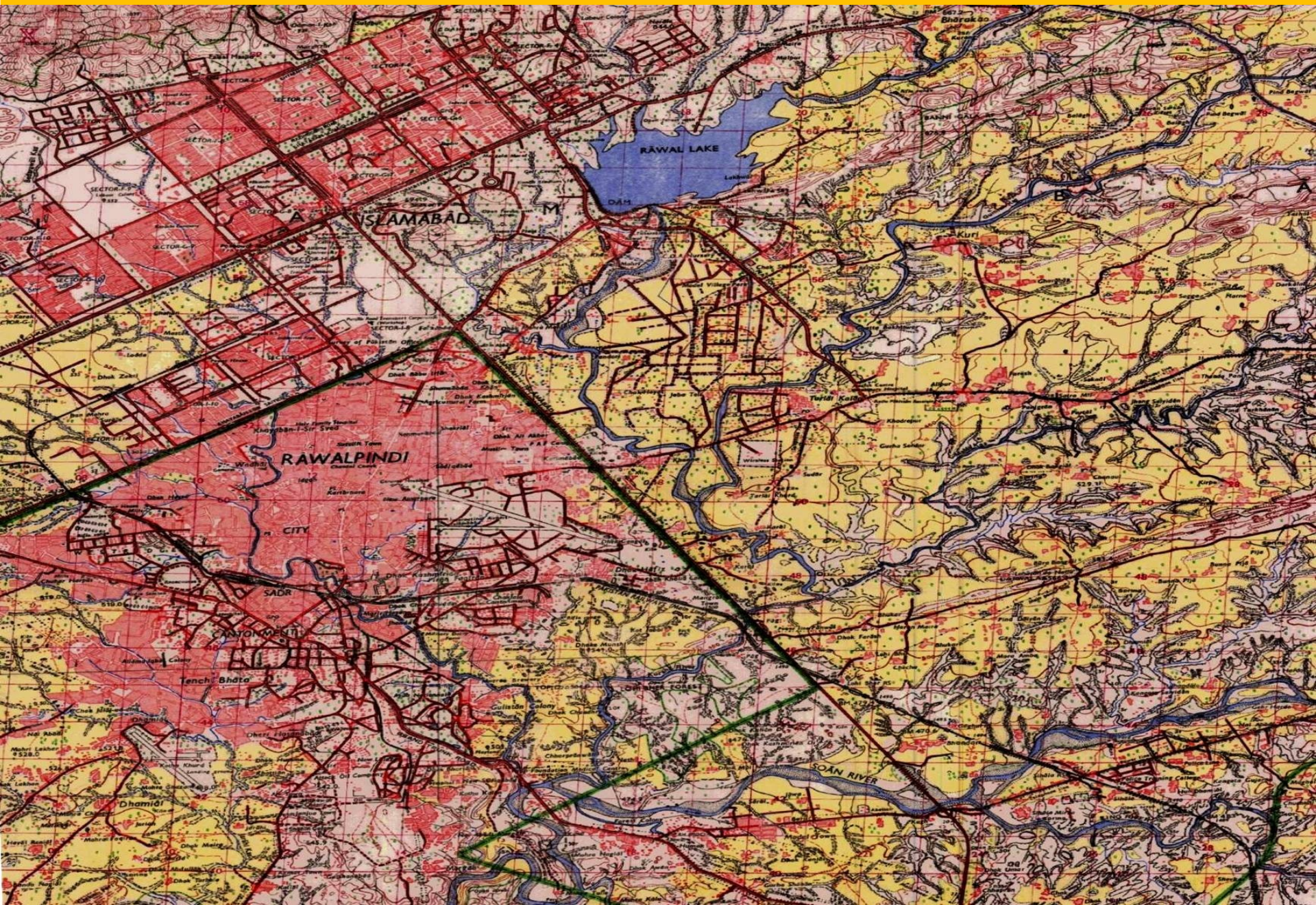
*"Through route" plan: loosely, a city plan; selection of towns and cities contingent on their offering complex traffic problems where 1:250,000, scale map cannot show required information.

Printed by the U. S. Army Topographic Command 150117

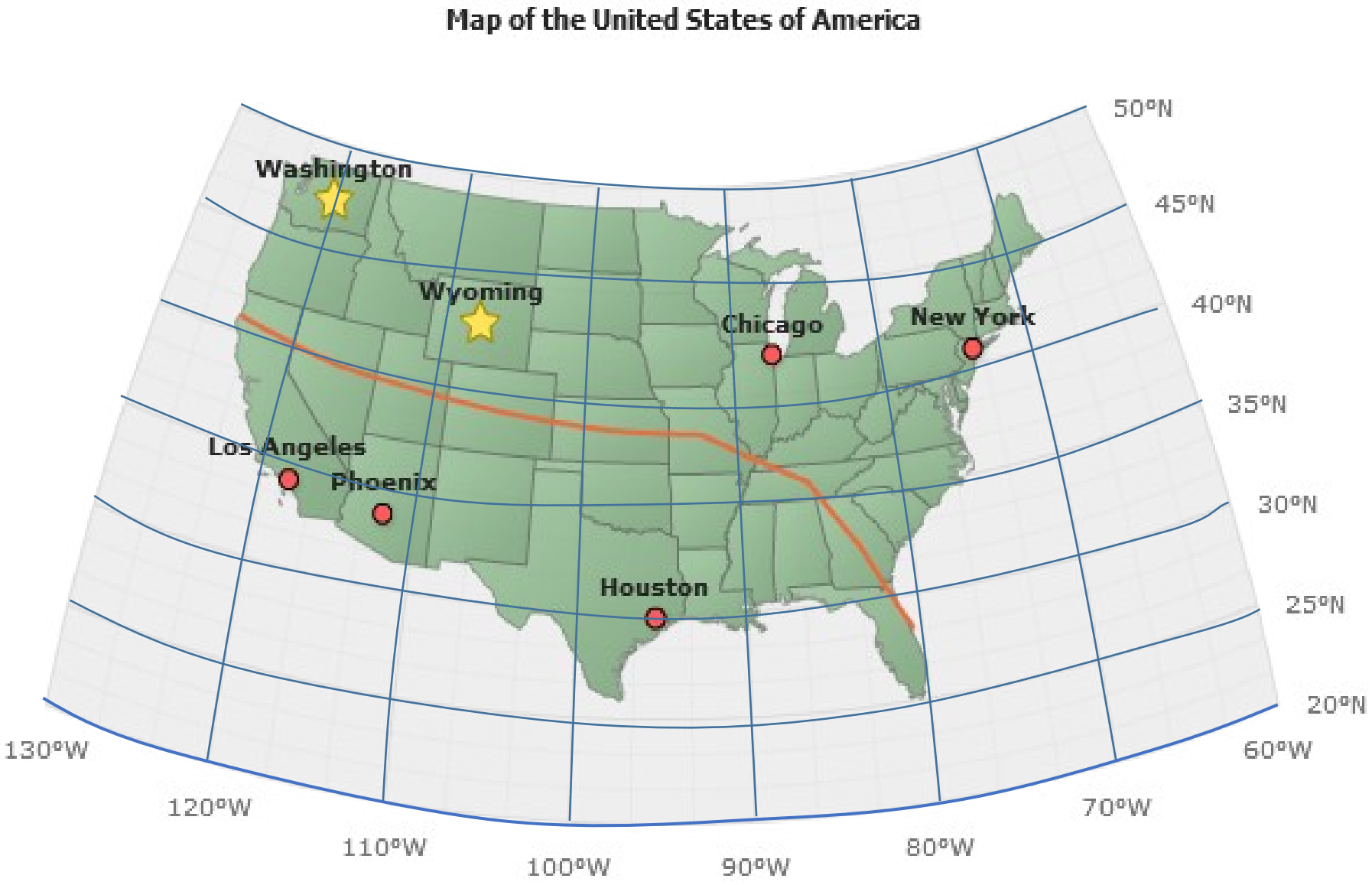
# MAP GRID SYSTEM

- It comprises of a series of squares formed by vertical and horizontal lines called ‘eastings’ and ‘northings’ which are serially numbered e.g. 31,32,33 etc.
- Grid System is used to identify the position of any ground point or object on map by a four figure or six figure reference. On small scale map (1:million scale etc.) the series of squares in formed by longitudes and latitudes called graticules.

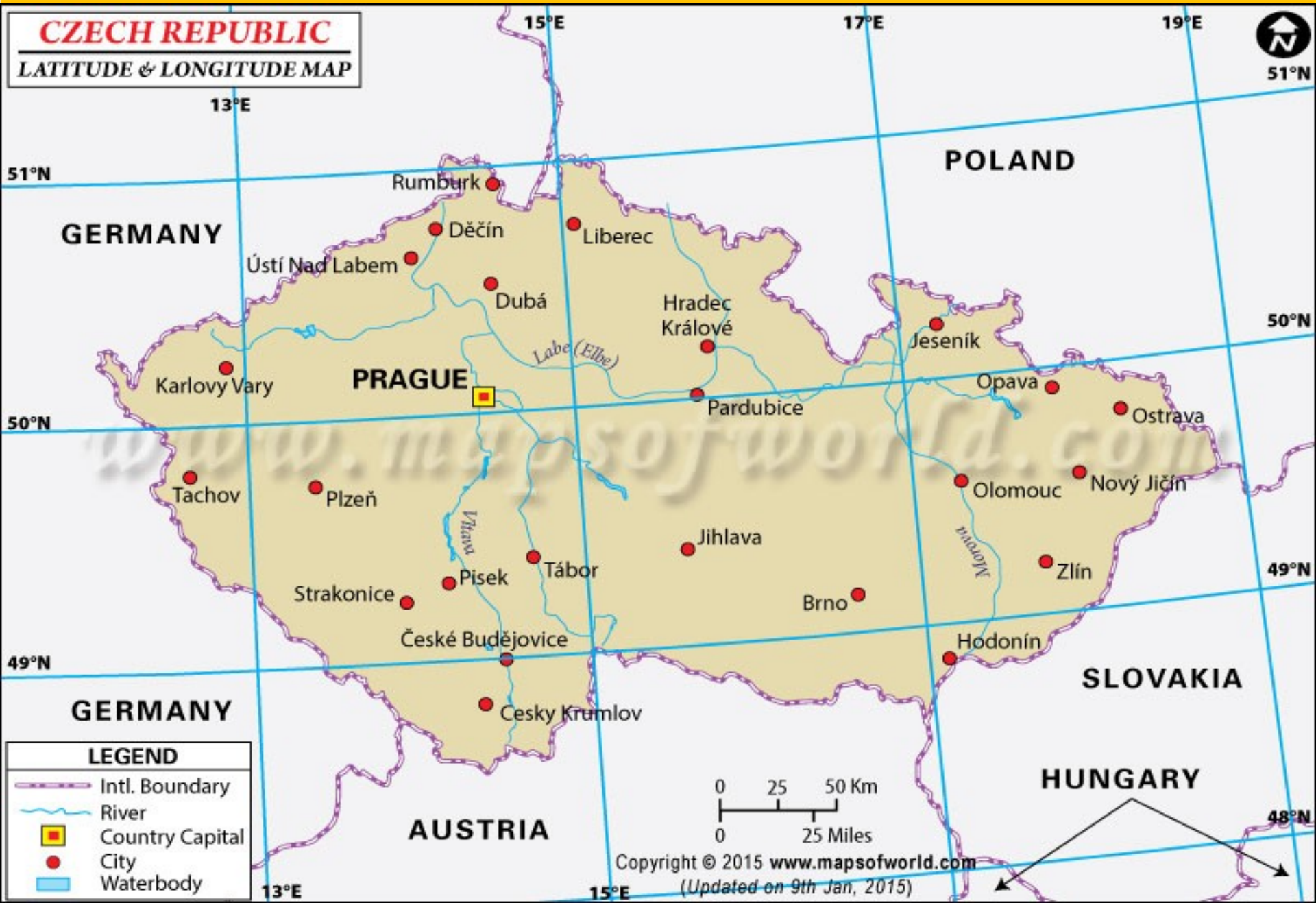
# ISLAMABAD MAP



# MAP GRATICULES



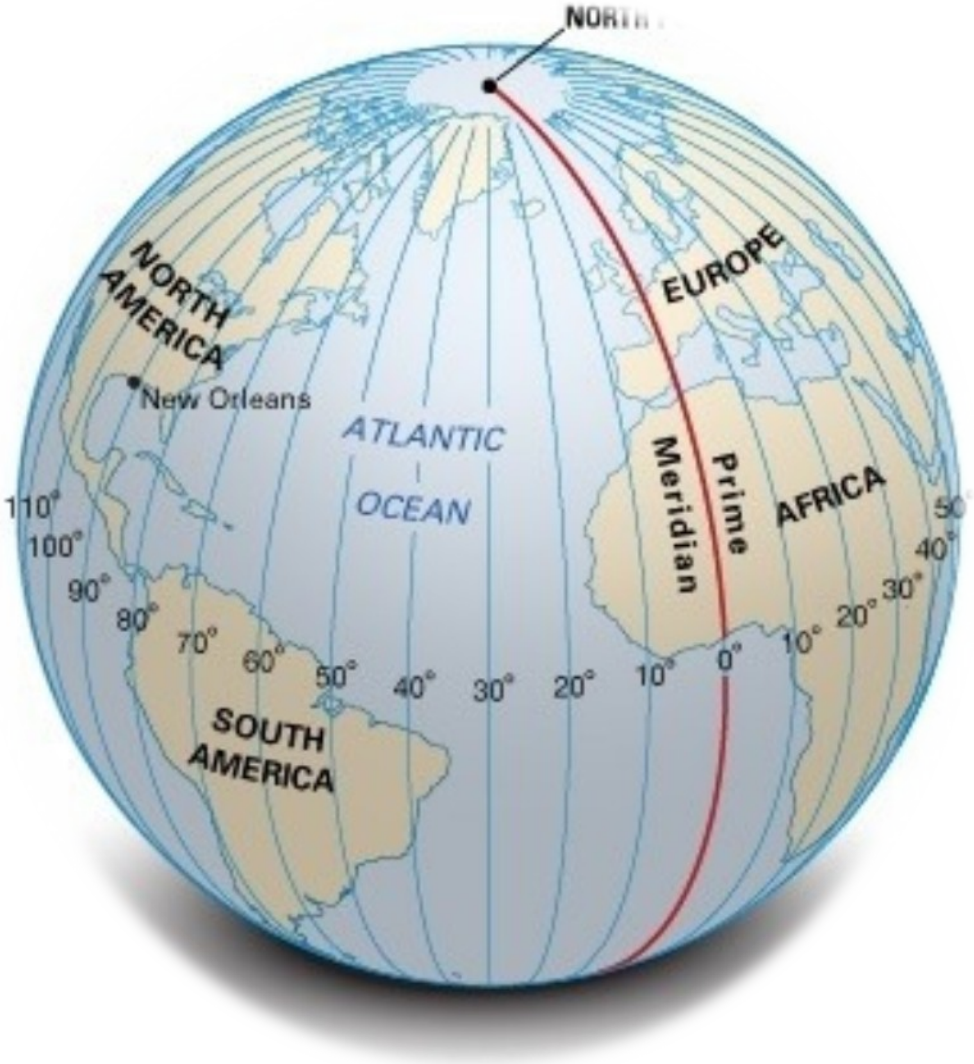
# MAP GRATICULES



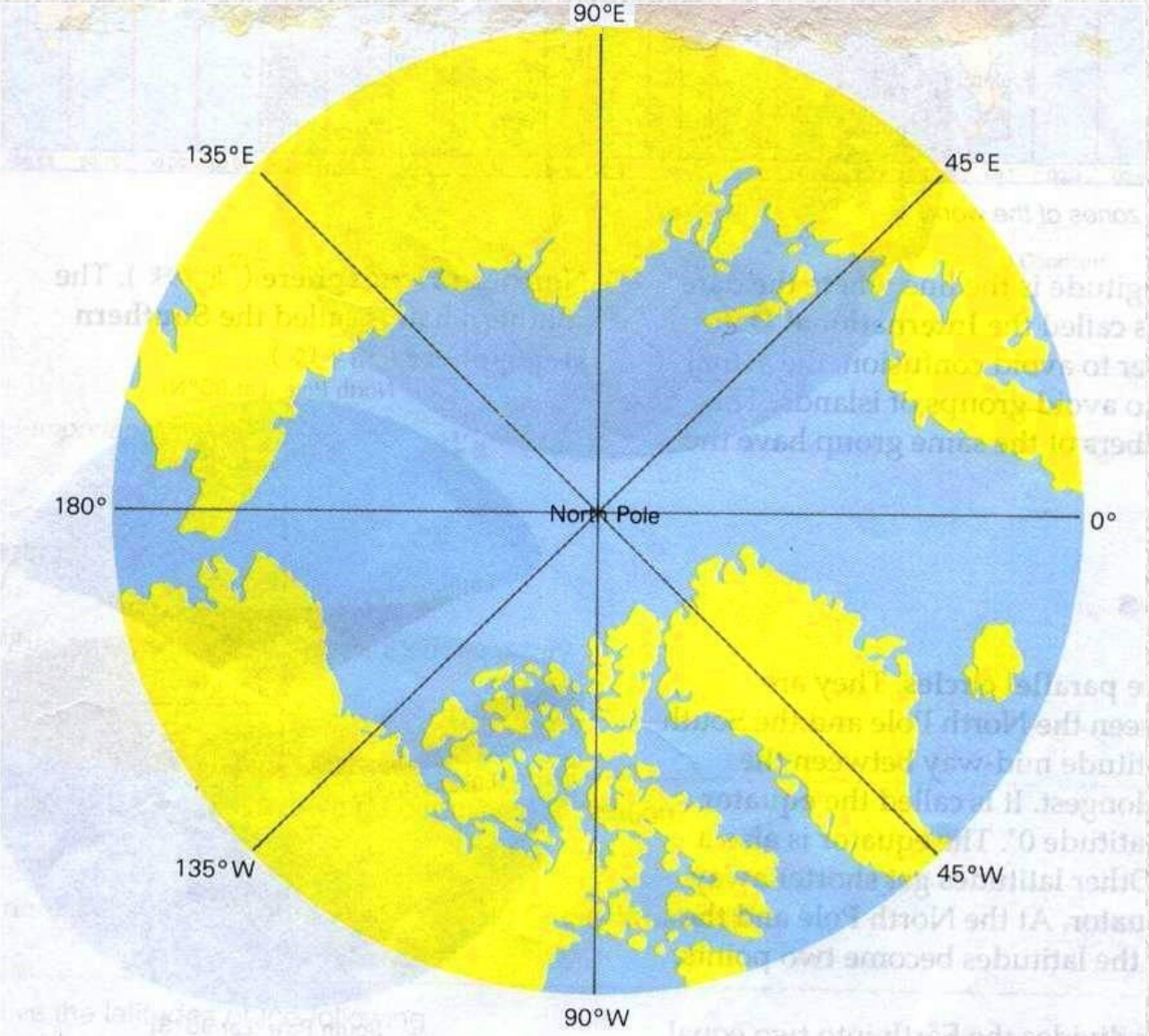
# GEOGRAPHIC COORDINATE SYSTEM

- The location of a point can also be indicated anywhere on the surface of Globe by its Longitude and Latitude. Longitudes are vertical lines joining the two poles of the Globe. These are numbered 0˚- 180˚ East and 0˚- 180˚ West of the prime meridian passing through Green Wich.
- Latitude are horizontal lines running parallel to Equator and these are numbered 0˚- 90˚ North and 0˚- 90˚ South of Equator. On small scale maps we do not have grid lines and position of a location is indicated in terms of Longitude and Latitude.

# LONGITUDES


NORTH
NORTH AMERICA
New Orleans
ATLANTIC
OCEAN
EUROPE
Prime
Meridian
AFRICA
SOUTH AMERICA
110°
100°
90°
80°
70°
60°
50°
40°
30°
20°
10°
0°
10°
20°
30°
40°

# NUMBERING OF LONGITUDES

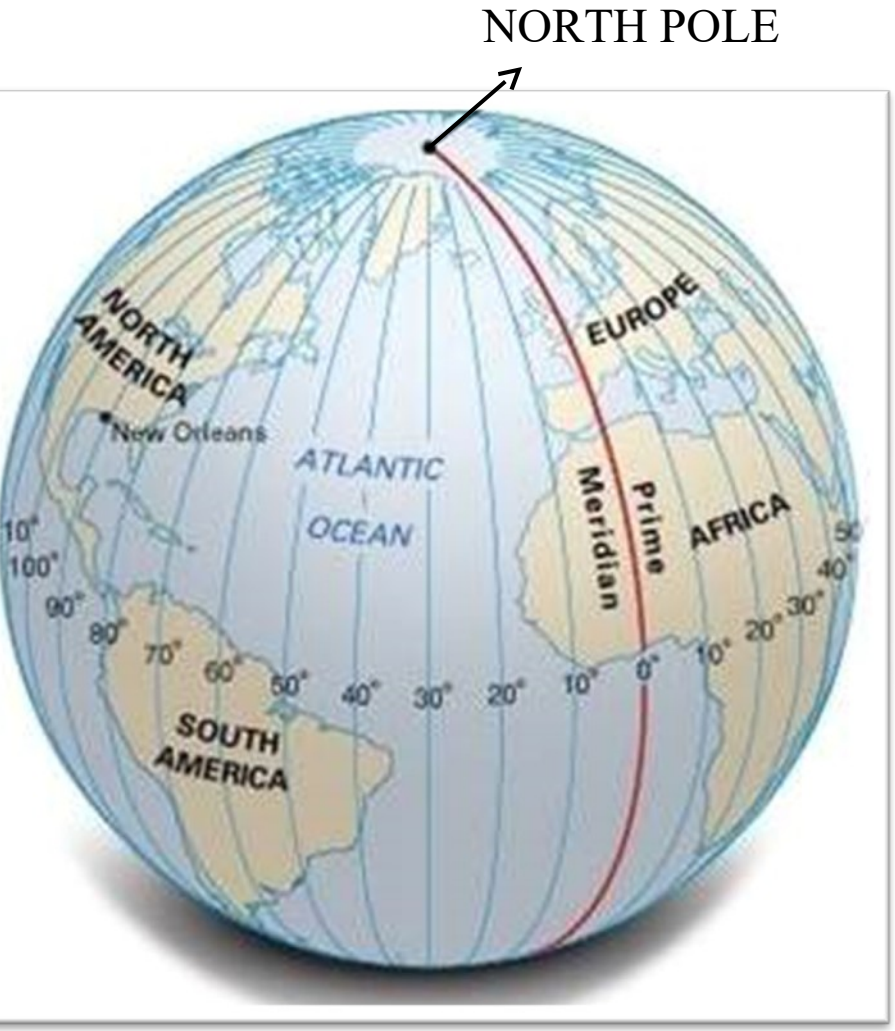



*Numbering of longitudes*

# LATITUDES

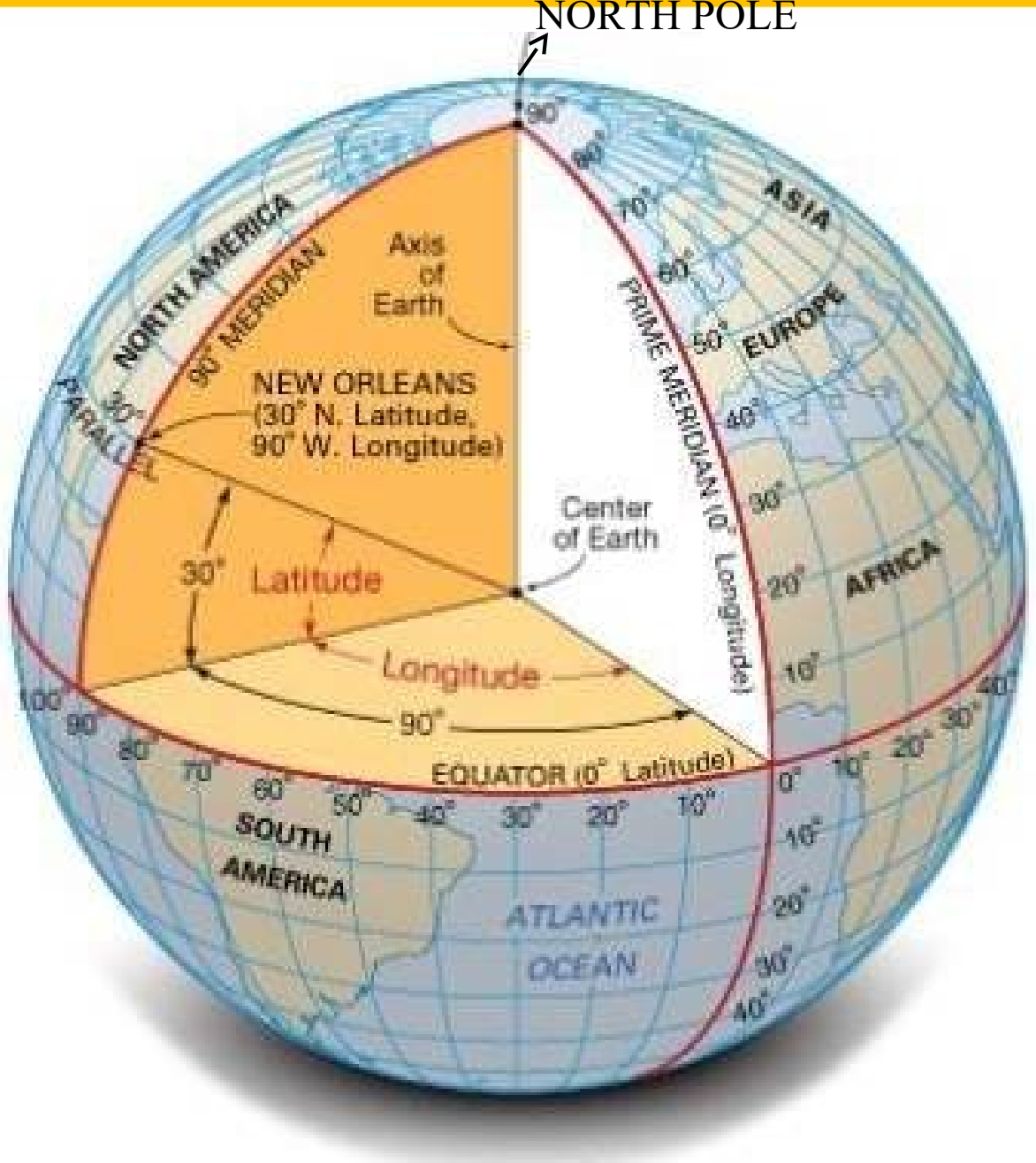


# LONGITUDES AND LATITUDES






# LONGITUDES AND LATITUDES



# MEASURMENT OF LATITUDES

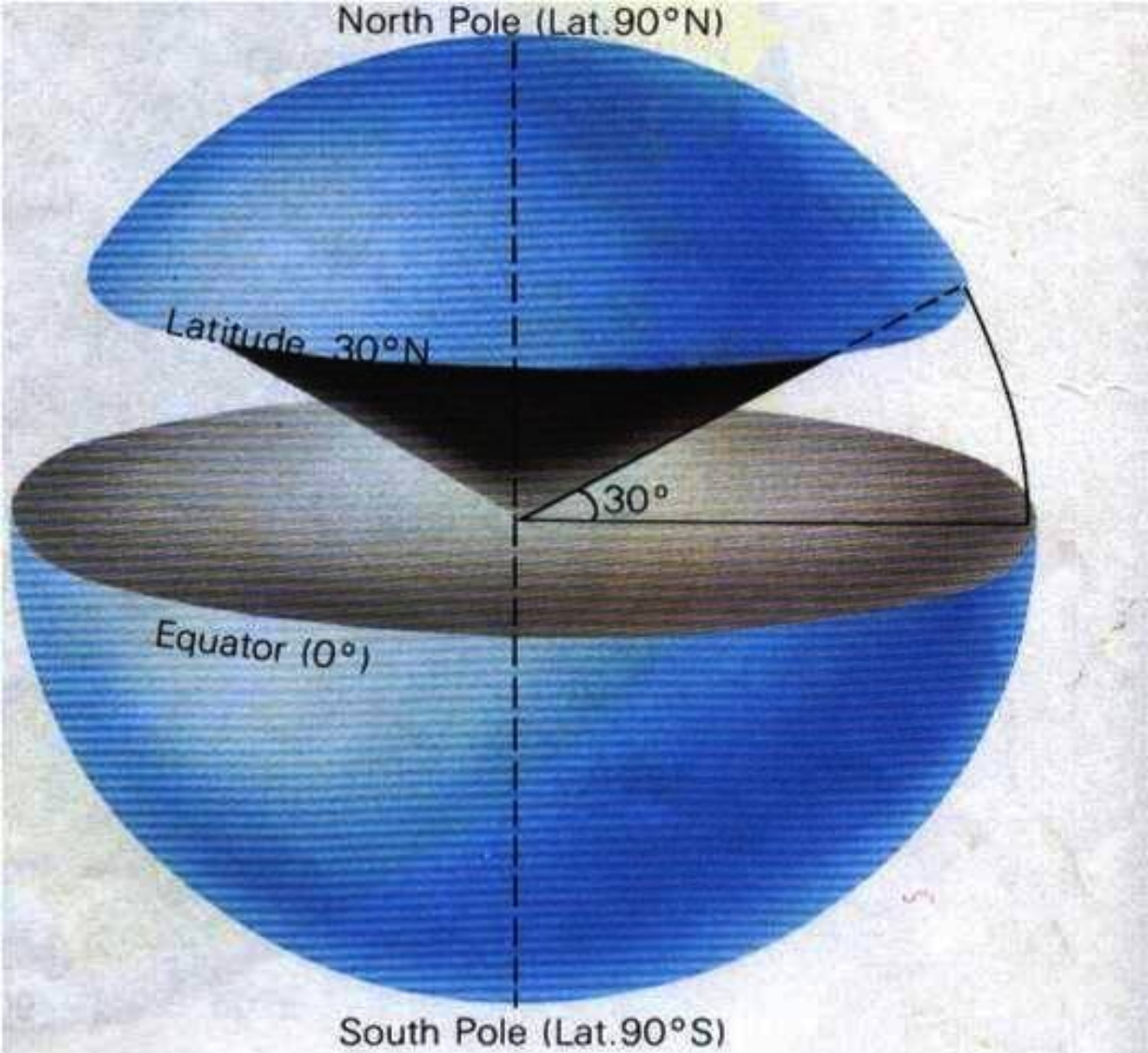
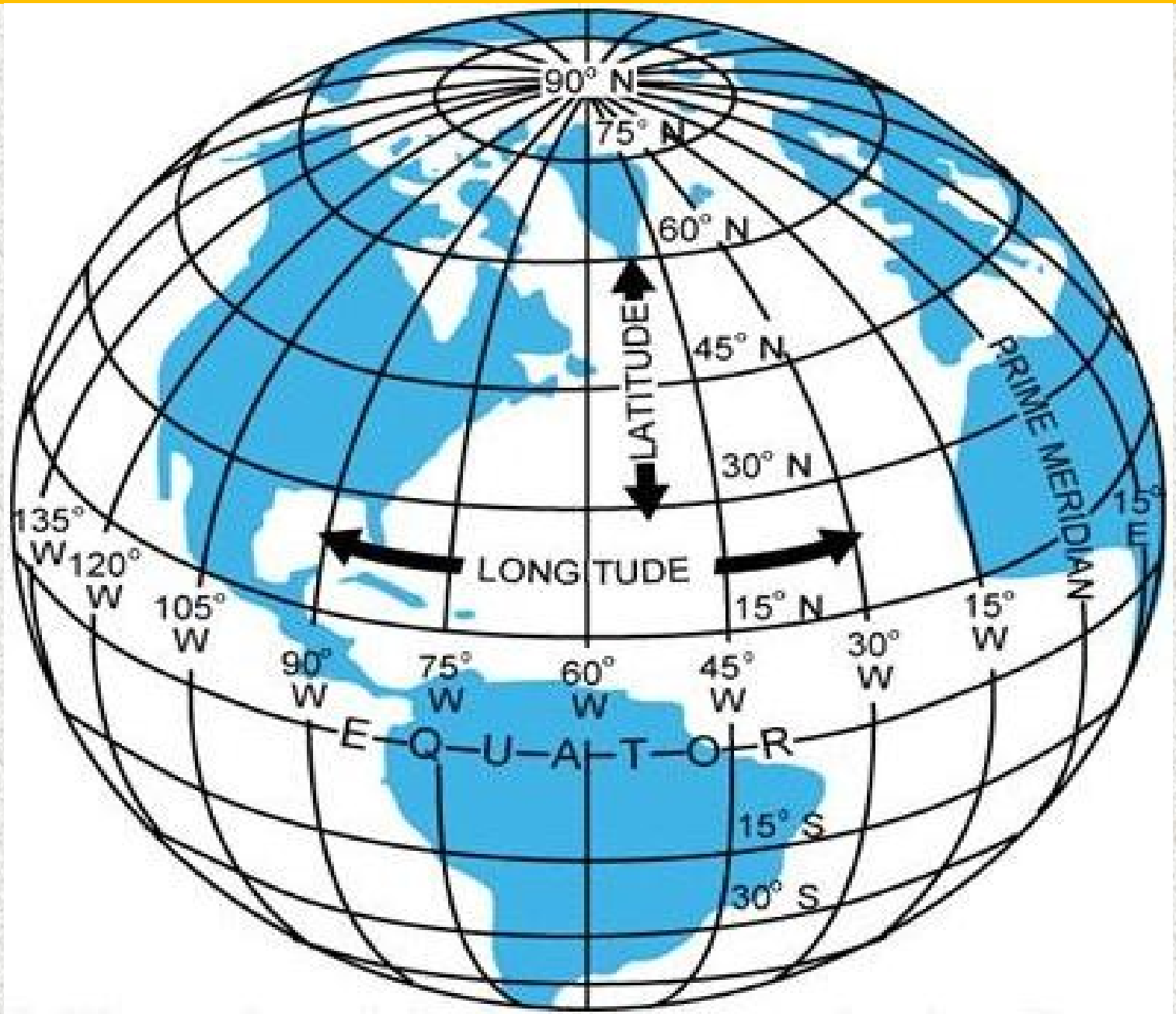
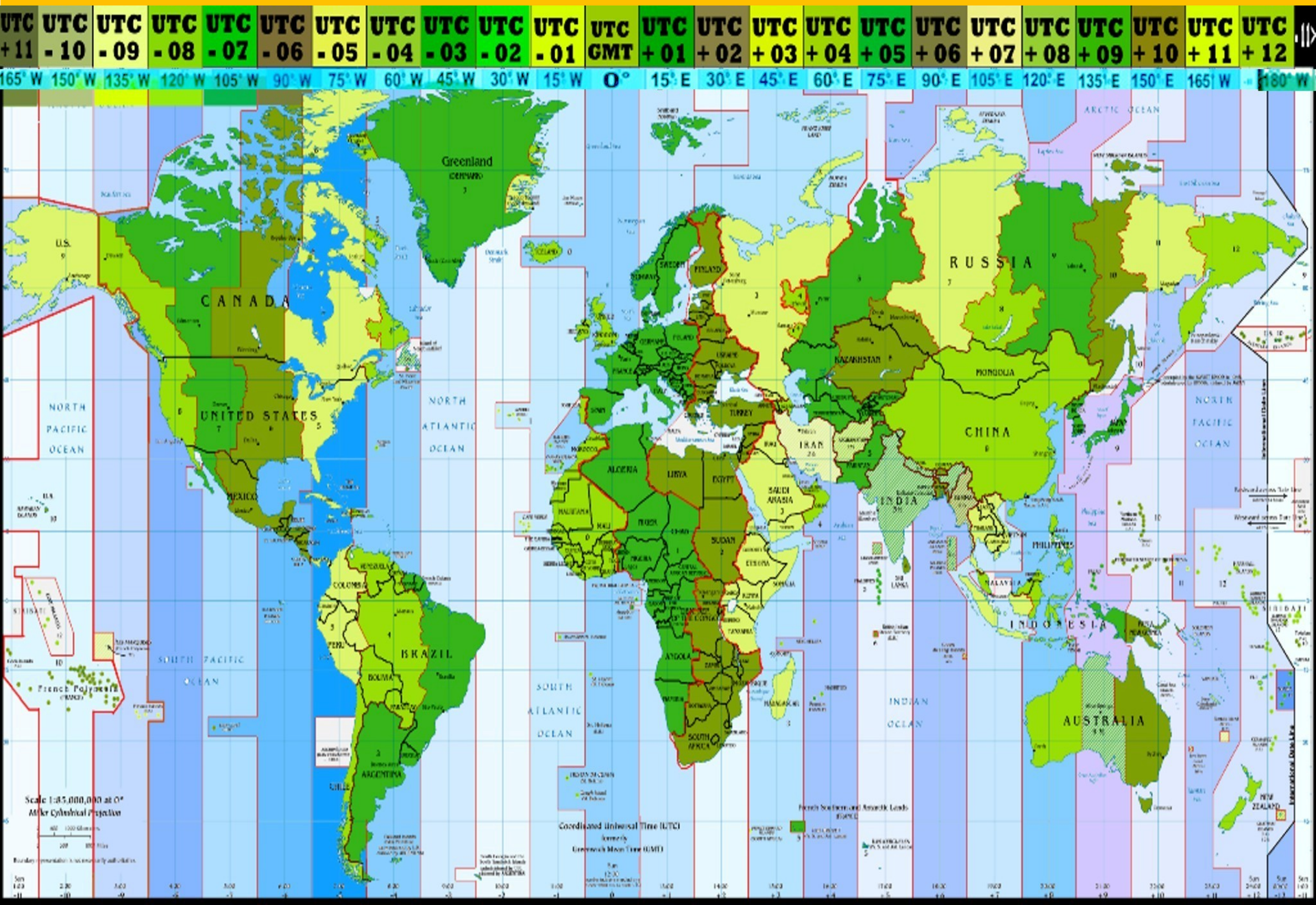



*Fig. 5 How latitudes are measured*

# MEASURMENT OF LATITUDES LONGITUDES



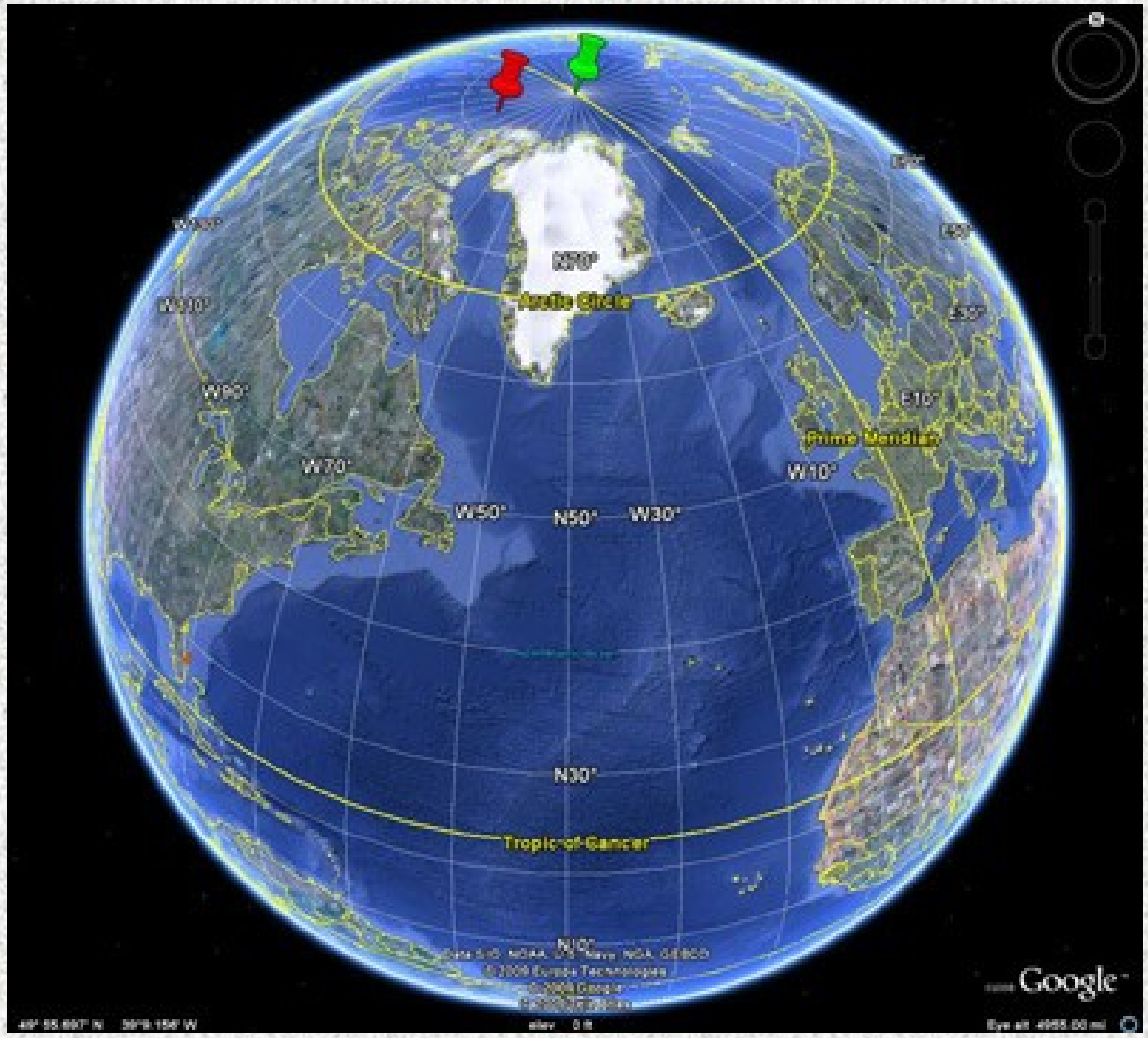
# TIME ZONES OF THE WORLD



# TYPES OF NORTH

- True North — The direction of north pole from observer.
- Magnetic North — It is the direction in which the compass needle points towards the magnetic pole.
- Grid North — The direction in which the grid lines point towards the top of map.

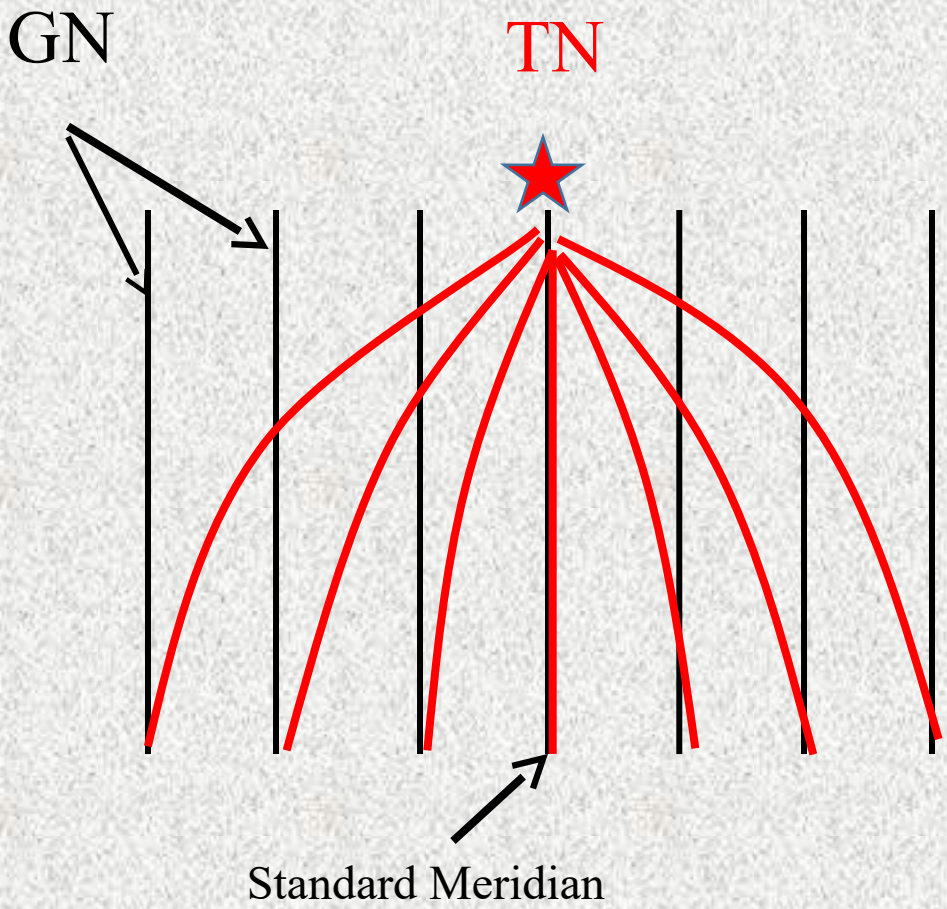
# TYPES OF NORTH



# TYPES OF NORTH

GRID NORTH

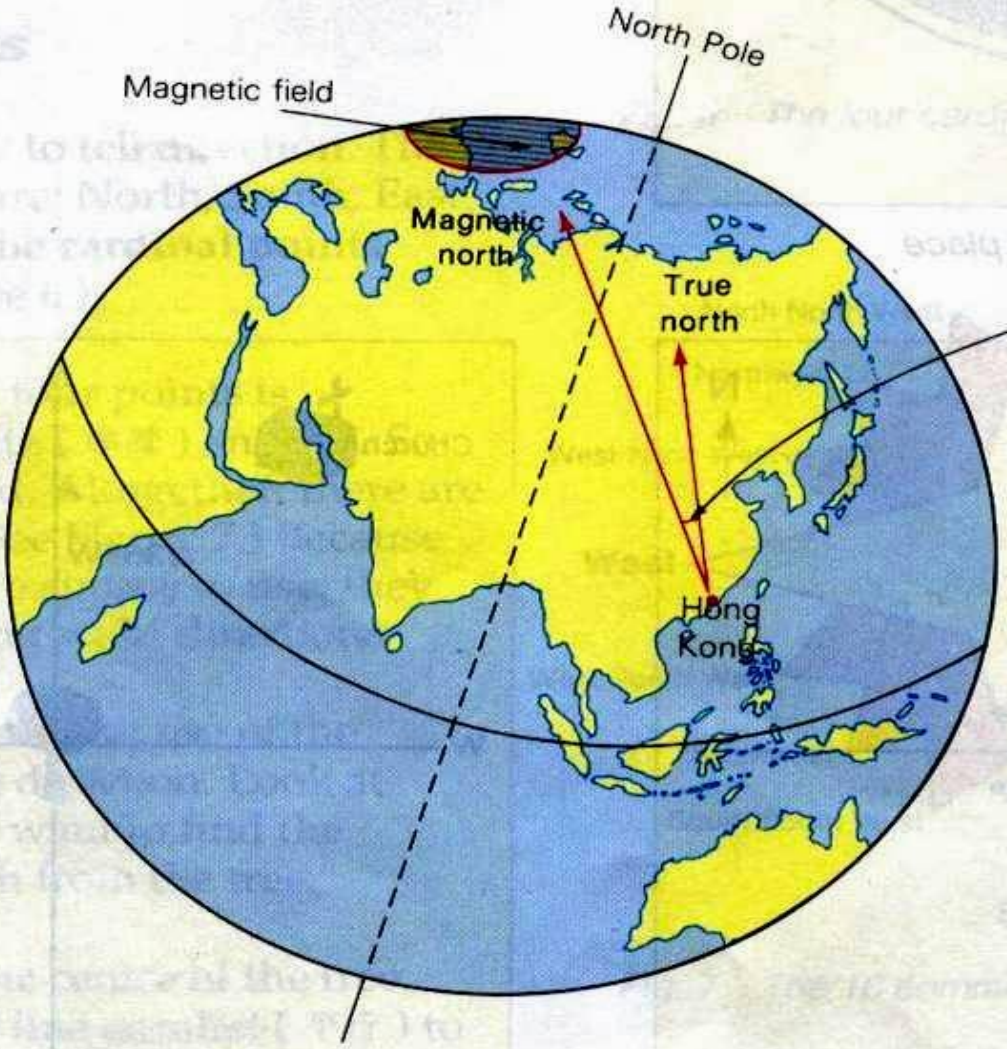
TRUE NORTH



# MAGNETIC DECLINATION/VARIATION

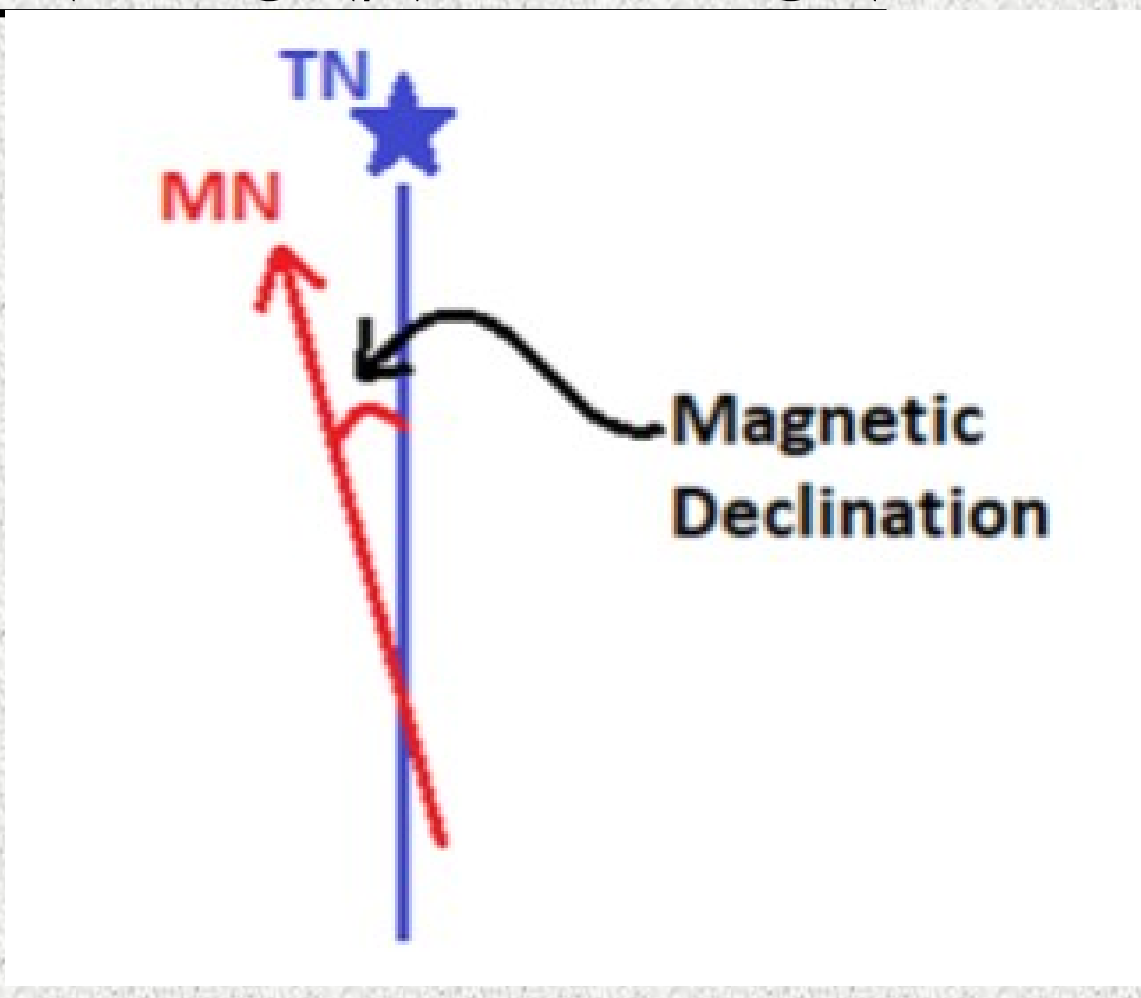
- The true north line and magnetic north line at a place seldom coincide with each other due to which there is a difference between them.
- This difference is called magnetic declination.

# MAGNETIC DECLINATION



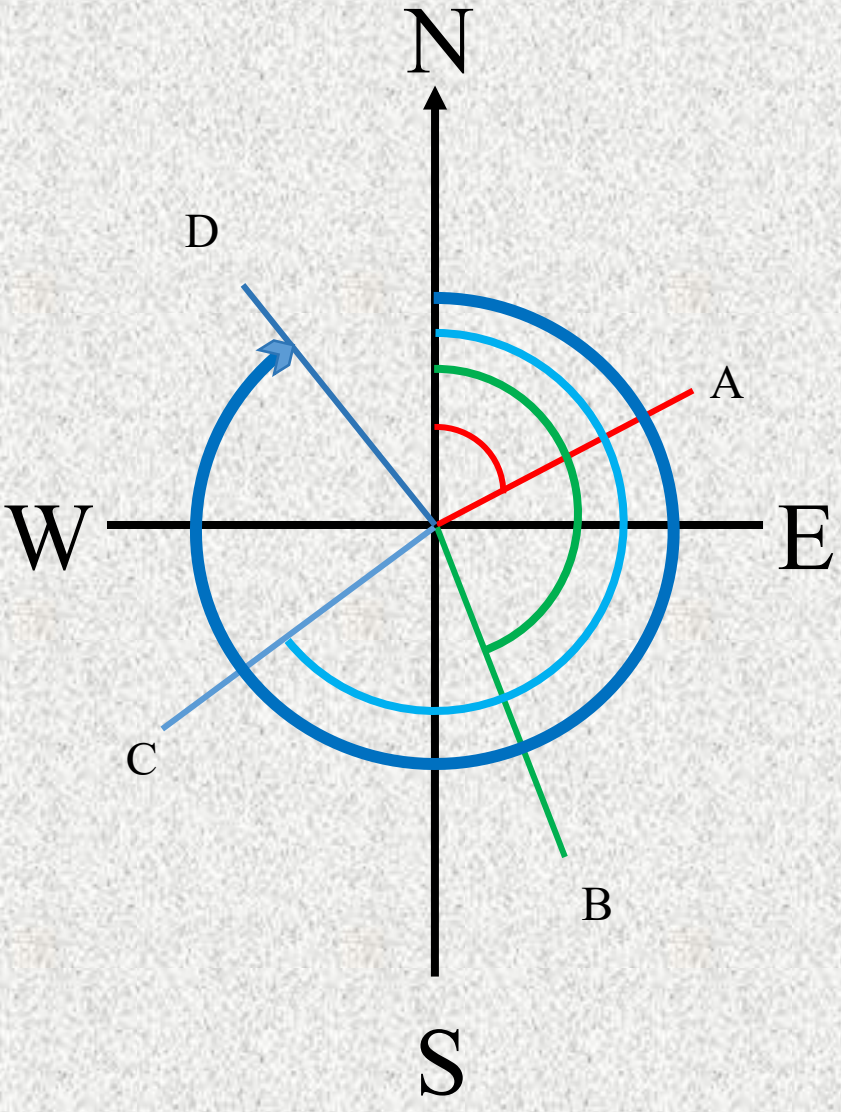

Magnetic north and true north

# MAGNETIC DECLINATION/VARIATION



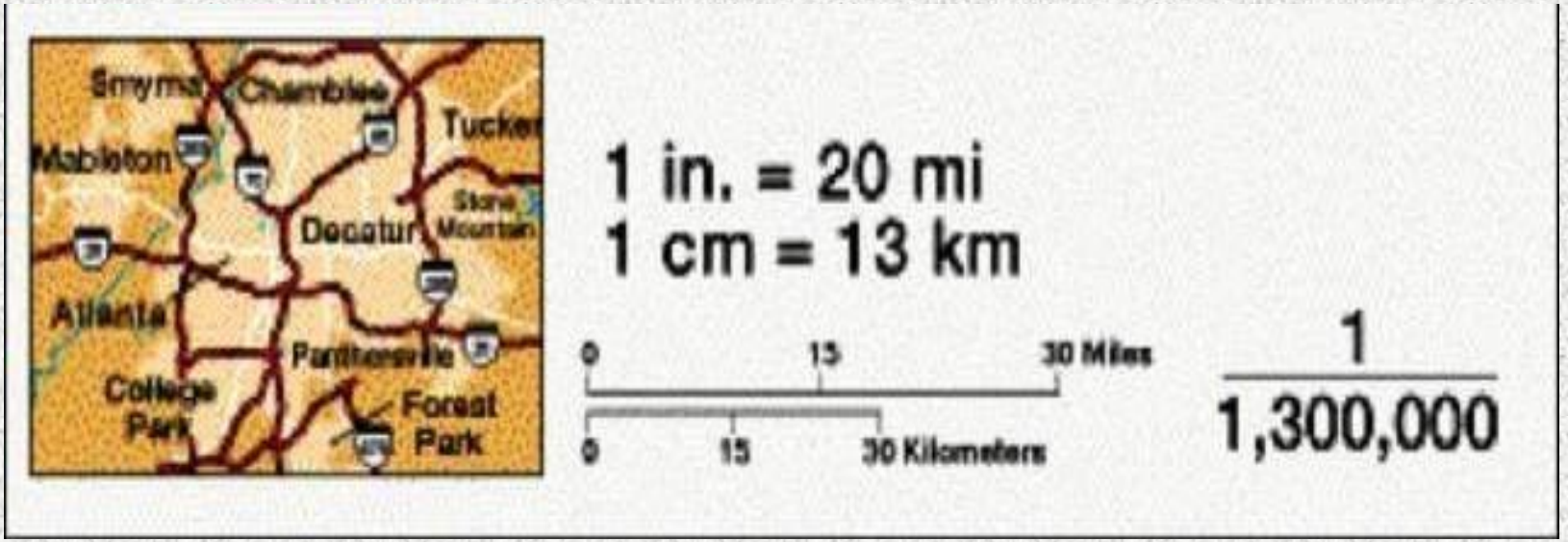
# BEARING

Angle measured clockwise from a certain fixed North-South Line to an object.



# SCALE

- The proportion which the distance between any two points on map bears to the horizontal distance between the same two points on the ground.



# METHODS OF EXPRESSING SCALE

- **IN WORDS**

  Words explain the distance on map that represents a certain distance on ground.

  e.g. 1 Inch=1 Mile etc.

- **BY REPRESENTATIVE FRACTION**

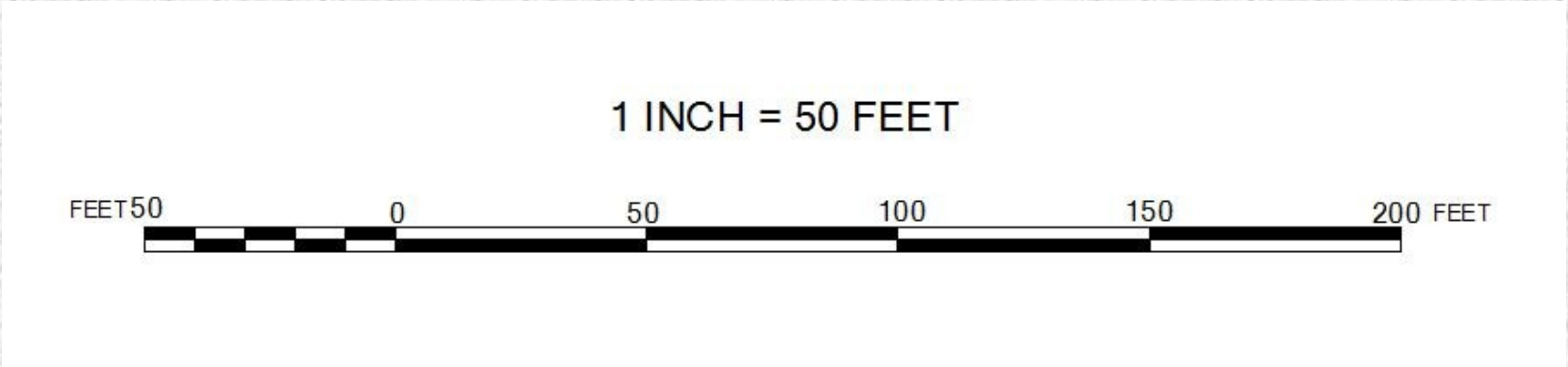
  The distance on map is represented by a fraction of corresponding distance on ground.

  e.g. 1:50,000 , 1/10,000 etc.

# METHODS OF EXPRESSING SCALE

- **BY SCALE LINE**

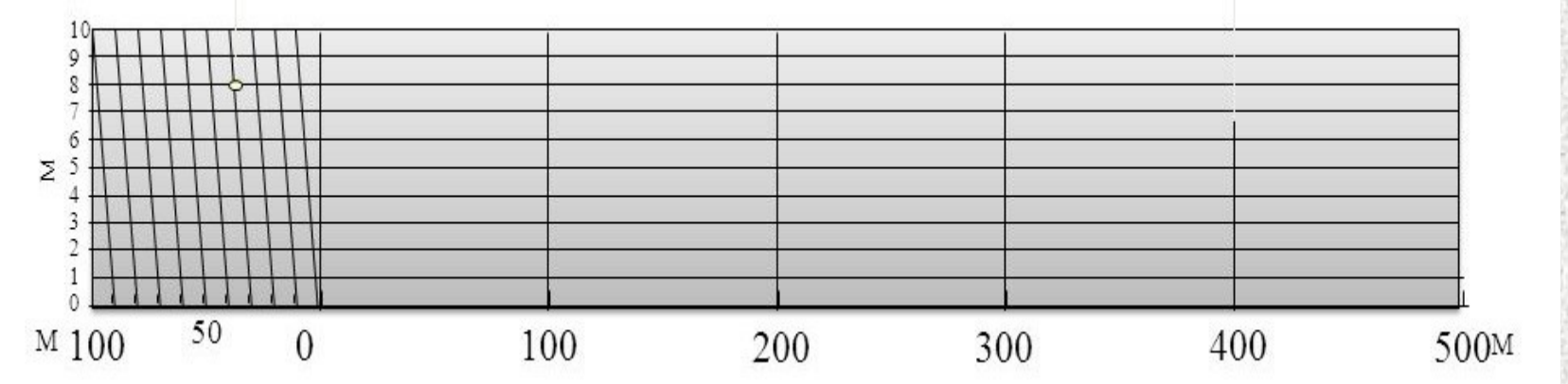
  By drawing a scale line showing the digits or parts for measuring distance on the map.



# METHODS OF EXPRESSING SCALE

- **DIAGONAL SCALE**

  By drawing this type of scale line it is possible to measure distances accurately to the fraction of a secondary division of a scale line to a required standard.



# Questions